

ADAMTS13 Assay for Thrombotic Thrombocytopenic Purpura (TTP) - Medical Clinical Policy Bulletins

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Policy

Scope of Policy

This Clinical Policy Bulletin addresses ADAMTS13 assay for thrombotic thrombocytopenic purpura (TTP).

1. Medical Necessity

Aetna considers the ADAMTS13 assay medically necessary for assessing prognosis in persons with thrombotic thrombocytopenic purpura (TTP).

2. Experimental, Investigational, or Unproven

Aetna considers all of the following as experimental, investigational, or unproven:

1. ADAMTS13 assay for the following indications (not an all-inclusive list) because of insufficient evidence of its clinical utility for these indications:
1. Biomarker for delayed cerebral ischemia after subarachnoid hemorrhage
2. Biomarker for disease severity and risk of micro-thrombosis in individuals with coronavirus disease-2019 (COVID-19)
3. Biomarker for early detection of hepatocellular carcinoma
4. Biomarker for monitoring progression of acute myocardial infarction
5. Biomarker for sorafenib treatment efficiency in individuals with hepato-cellular carcinoma
6. Biomarker for treatment response in individuals with hepatocellular carcinoma before the initiation of hepatic arterial infusion chemotherapy
7. Determining the risk of endometriosis
8. Diagnosis and monitoring of diabetic retinopathy
9. Diagnosis and prognosis of portal vein thrombosis in individuals with liver cirrhosis
10. Diagnosis and the therapeutic monitoring of individuals with sepsis associated thrombotic microangiopathy

11. Diagnosis of acute cholangitis
12. Diagnosis of acute myelogenous leukemia
13. Diagnosis of acute pancreatitis
14. Diagnosis of arterial thrombosis
15. Diagnosis of cerebral infarction
16. Diagnosis of systemic sclerosis (scleroderma)
17. Disseminated intravascular coagulation
18. Evaluation and prognosis of cerebral small vessel disease
19. Evaluation of disease activity in inflammatory bowel diseases
20. Hemolytic uremic syndrome (HUS)
21. Individuals with type 1 Von Willebrand disease who plan on undergoing major surgery (e.g., total hip replacement)
22. Ischemic complications of malignant hypertension
23. Monitoring of liver diseases
24. Monitoring of pregnant COVID-19 women
25. Monitoring of renal function following kidney transplantation
26. Monitoring the development and progression of obstructive sleep apnea
27. Prediction of acute kidney injury in individuals with COVID-19
28. Prediction of adverse cardiovascular outcomes in individuals undergoing percutaneous coronary intervention
29. Prediction of adverse outcomes in individuals with non-valvular atrial fibrillation
30. Prediction of excessive post-operative drainage after coronary artery bypass grafting
31. Prediction of hepatocellular carcinoma development
32. Prediction of long-term HBV e antigen (HBeAg) sero-conversion in individuals with chronic hepatitis B
33. Prediction of lung cancer development
34. Prediction of outcomes following subarachnoid hemorrhage
35. Prediction of recurrence of atrial fibrillation
36. Predicting of recurrence of venous thromboembolism
37. Prediction of relapse and survival following allogeneic hematopoietic stem cell transplantation from unrelated donors
38. Prediction of response to recanalization therapies in acute ischemic stroke
39. Prediction of severe outcomes in individuals hospitalized with COVID-19
40. Prediction of survival in colorectal cancer
41. Prediction of thrombotic risk in persons with systemic lupus erythematosus
42. Pre-eclampsia
43. Prognostic marker of melanoma
44. Prognosis of traumatic brain injury
45. Risk factor for intracranial aneurysm
46. Risk factor for thrombosis in pediatric congenital heart disease;
2. ADAMTS13 mutation testing for diagnosis of non-cirrhotic portal hypertension because of

insufficient evidence;

3. Use of the ratio of von Willebrand factor antigen to ADAMTS13 activity as a prognostic biomarker in acute-on-chronic liver failure.

CPT Codes / HCPCS Codes / ICD-10 Codes	
Code	Code Description
CPT codes covered if selection criteria are met:	
85397	Coagulation and fibrinolysis, functional activity, not otherwise specified (eg, ADAMTS-13), each analyte
Other CPT codes related to the CPB:	
38204 - 38205	Allogeneic hematopoietic stem cell transplantation
38208 - 38215	
38240	Hematopoietic progenitor cell (HPC); allogeneic transplantation per donor
96413	Chemotherapy administration, intravenous infusion technique; up to 1 hour, single or initial substance/drug
96415	Chemotherapy administration, intravenous infusion technique; each additional hour (List separately in addition to code for primary procedure)
96416	Chemotherapy administration, intravenous infusion technique; initiation of prolonged chemotherapy infusion (more than 8 hours), requiring use of a portable or implantable pump
96417	Chemotherapy administration, intravenous infusion technique; each additional sequential infusion (different substance/drug), up to 1 hour (List separately in addition to code for primary procedure)
ICD-10 codes covered if selection criteria are met:	
M31.10	Thrombotic microangiopathy, unspecified
M31.11	Hematopoietic stem cell transplantation-associated thrombotic microangiopathy [HSCT-TMA]
M31.19	Other thrombotic microangiopathy
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	
A41.9 [reported with M31.10 - M31.19]	Sepsis, unspecified [sepsis associated with thrombotic microangiopathy]

Code	Code Description
B18.0	Chronic viral hepatitis B with delta-agent
B18.1	Chronic viral hepatitis B without delta-agent
C18.0 - C20	Malignant neoplasm of colon, rectosigmoid junction and rectum [for prediction of survival in colorectal cancer]
C22.0	Liver cell carcinoma
C43.0 - C43.9	Malignant melanoma of skin [for use as a prognostic marker]
C92.10 - C92.92	Acute myeloid leukemia
D59.30 - D59.39	Hemolytic-uremic syndrome
D68.01	Von Willebrand disease, type 1
• E08.311 - E08.3599, • E09.311 - E09.3599, • E10.311 - E10.3599, • E11.311 - E11.3599, • E13.311 - E13.3599	Diabetic retinopathy [diagnosis and monitoring of]
G47.33	Obstructive sleep apnea (adult) (pediatric)
I10 - I16.9	Hypertensive diseases
I21.01 - I21.9	Acute myocardial infarction
I24.0	Acute coronary thrombosis not resulting in myocardial infarction
• I48.0 - I48.21, • I48.91	Atrial fibrillation
I60.00 - I60.9	Nontraumatic subarachnoid hemorrhage [for prediction of outcomes following subarachnoid hemorrhage]
I63.00 - I63.9	Cerebral infarction [prediction of response to recanalization therapies in acute ischemic stroke]
I65.01 - I66.9	Occlusion and stenosis of precerebral and cerebral arteries
I67.0 - I67.9	Other cerebrovascular diseases [Cerebral small vessel disease]
I74.01 - I74.9	Arterial embolism and thrombosis
I81	Portal vein thrombosis
I82.0 - I82.91	Other venous embolism and thrombosis [for prediction of recurrence of venous thromboembolism]
J96.00 - J96.92	Respiratory failure, not elsewhere classified 9 [ratio of von Willebrand factor antigen to ADAMTS13 activity as a prognostic biomarker]

Code	Code Description
K50.00 - K50.919	Crohn's disease [regional enteritis] [for evaluation of disease activity in inflammatory bowel diseases]
K51.00 - K51.919	Ulcerative colitis [for evaluation of disease activity in inflammatory bowel diseases]
K70.0 - K77	Diseases of liver
K83.01 - K83.09	Cholangitis [for diagnosis of]
K85.00 - K85.92	Acute pancreatitis [for diagnosis of]
M32.10 - M32.9	Systemic lupus erythematosus [prediction of thrombotic risk in persons with systemic lupus erythematosus]
M34.0 - M34.9	Systemic sclerosis [scleroderma]
N17.9	Acute kidney failure, unspecified
N80.00 - N80.D9	Endometriosis [determining the risk]
010.911 - 011.9 014.00 - 016.9	Hypertension, pre-eclampsia and eclampsia in pregnancy, childbirth and the puerperium
098.519	Other viral diseases complicating pregnancy, unspecified trimester [monitoring of pregnant COVID-19 women]
Q24.9	Congenital malformation of heart, unspecified
R65.10 - R65.21 [reported with M31.1]	Symptoms and signs specifically associated with systemic inflammation and infection [sepsis associated with thrombotic microangiopathy]
S06.0X0A - S06.A1XS	Traumatic brain injury
S06.9x0A - S06.9x9S	Unspecified intracranial injury [for prognosis of traumatic brain injury]
U07.1	COVID-19
Z12.2	Encounter for screening for malignant neoplasm of respiratory organs
Z94.0	Kidney transplant status [monitoring of renal function following kidney transplantation]
Z94.84	Stem cells transplant status [for prediction of relapse and survival following allogeneic hematopoietic stem cell transplantation from unrelated donors]
Z98.890 - Z98.891	Other specified postprocedural states [excessive post-operative drainage after coronary artery bypass grafting]
ADAMTS13 mutation testing - no specific code:	
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	
K76.6	Portal hypertension [non-cirrhotic portal hypertension]

Background

ADAMTS13 (A Disintegrin-like And Metalloprotease with ThromboSpondin type 1 motif) is a multi-domain protease that limits platelet thrombo-genesis through the cleavage of von Willebrand factor.

Retrospective studies of patients with thrombotic micro-angiopathies have shown that a deficient activity of ADAMTS13 in plasma is involved in thrombotic thrombocytopenic purpura (TTP) but not in hemolytic-uremic syndrome (HUS). It has been demonstrated that patients with inherited TTP have severe ADAMTS13 deficiency. However, patients with acquired TTP present with clinical and laboratory heterogeneity, and there are unequivocal cases of acquired TTP with measurable plasma levels of ADAMTS13. This heterogeneity poses a challenge for understanding the pathogenesis of TTP and selecting appropriate therapies (Mannucci and Peyvandi, 2007). The ADAMTS13 assay has been proposed by some to distinguish chronic recurring TTP (secondary to the presence of ADAMTS13 inhibitor) and HUS.

In a multi-center prospective cohort study (n = 111), Veyradier et al (2001) reported that most patients diagnosed with HUS had normal plasma levels of ADAMTS13, even though a few of them (14 %) had reduced or even undetectable levels. However, the high diagnostic value of finding severe ADAMTS13 deficiency in TTP was subsequently challenged by other studies (Loof et al, 2001; Moore et al, 2001; Mori et al, 2002; Remuzzi et al, 2002; Coppo et al, 2004; Peyvandi et al, 2004); 2 of which involved prospectively recruited cohorts (Vesely et al, 2003; Zheng et al, 2004). Some of these studies also investigated patients with the form of HUS preceded by hemorrhagic colitis that occurs typically in children, or the atypical form that occurs more frequently in adults. Atypical HUS is sometime indistinguishable from TTP unless signs and symptoms of severe renal impairment are prominent (Mannucci and Peyvandi, 2007). The majority of these studies (Veyradier et al, 2001; Remuzzi et al, 2002; Vesely et al, 2003; Coppo et al, 2004) confirmed that ADAMTS13 is normal or only slightly decreased in typical colitis-associated HUS. However, in a few patients diagnosed with atypical HUS, ADAMTS13 was as severely deficient as in TTP (Loof et al, 2001; Veyradier et al, 2001; Remuzzi et al, 2002). Studies showing that protease activity was also reduced in plasma in an array of clinical conditions other than TTP (e.g., spanning from various thrombocytopenic disorders to disseminated intravascular coagulation, sepsis, the neonatal and post-operative period, liver cirrhosis and chronic inflammation) further challenge the paradigm that ADAMTS13 deficiency is a specific diagnostic beacon of TTP (Moore et al, 2001; Mannucci et al, 2001; Bianchi et al, 2002). In these conditions, however, ADAMTS13 deficiency was usually moderate or mild (10 % to 40 % of normal plasma value).

Mannucci and Peyvandi's (2007) reviewed the literature on ADAMTS13 and stated that it is not necessary to assay ADAMTS13 to diagnose TTP in the acute phase of the disease. Patients presenting with normal or moderately reduced ADAMTS13 can still be appropriately diagnosed with TTP. Furthermore, the authors stated, "[t]he decision to implement plasma therapy (infusion in patients with inherited disease, exchange in acquired disease) does not warrant the availability of ADAMTS13 values in real time. Clinicians need to identify patients who are more likely to relapse and develop chronic recurrent TTP. Patients who present with undetectable ADAMTS13 activity and detectable anti-ADAMTS13 during the acute episode and/or during first remission are more likely to experience other episodes. Therefore, ADAMTS13 testing appears to be more helpful as an index of relapse than as an index of short-term outcomes (remission and mortality rates), but larger confirmatory studies are warranted."

It has been posited that knowledge of the likelihood of relapse would be useful in avoiding stressors that can induce TTP. It has been observed that TTP can occur (or recur) after pregnancy, infection, pancreatitis, and surgery; acute stresses, resulting perhaps in the release of inflammatory cytokines or other prothrombotic mediators, can trigger an initial or recurrent episode, perhaps by altering the balance between levels of von Willebrand factor and ADAMTS13 activity in a susceptible patient.

van den Born et al (2008) stated that thrombotic microangiopathy (TMA) observed in malignant hypertension is similar to that of TTP, which is associated with a deficiency of ADAMTS13, a von Willebrand factor (VWF)-cleaving protease that cleaves large prothrombogenic multimers. These researchers hypothesized that ADAMTS13 is deficient in malignant hypertension and that the severity of TMA is associated with decreased ADAMTS13 activity. They included 20 patients with malignant and 20 patients with severe

hypertension, and 20 matched normotensive individuals served as control subjects. Free hemoglobin, VWF, and active VWF were assessed to explore predictors of ADAMTS13 activity. Patients with malignant hypertension had lower ADAMTS13 activity (80 %; interquartile range [IQR]: 53 % to 130 %) compared with control subjects (99 % IQR: 82 % to 129 %; $p < 0.01$) but not compared with patients with severe hypertension ($p = 0.14$). ADAMTS13 activity negatively correlated with lactic dehydrogenase levels after logarithmic transformation ($r = -0.65$; $p < 0.001$) and was associated with platelet count ($r = 0.34$; $p = 0.04$) and the presence of schistocytes ($r = -0.37$; $p = 0.02$). Apart from the association with TMA, ADAMTS13 was inversely associated with creatinine ($r = -0.42$; $p = 0.008$). Increasing levels of VWF were associated with a decrease in ADAMTS13 activity ($r = -0.34$; $p = 0.03$). There was no significant association between ADAMTS13 activity and other parameters, including blood pressure. The authors concluded that ADAMTS13 is decreased in malignant hypertension and associated with the severity of TMA, likely because of the release of VWF after endothelium stimulation. A severe deficiency could not be demonstrated. They stated that more studies are needed to identify the role of ADAMTS13 in the TMA and ischemic complications of malignant hypertension.

Claus and colleagues (2009) measured VWF and related parameters as well as the protease activity regulating its biological activity in plasma of healthy controls and patients with different cause and severity of systemic inflammation to examine the effectiveness of the measures to detect highly prothrombotic states including TMA, one of the sequelae of sepsis. Plasma levels of VWF increased with increasing severity of systemic inflammation, probably due to activation of the endothelium. In parallel, the proteolytic activity of VWF inactivating protease, ADAMTS13, stepwise declined with the severity of inflammation, emphasizing the role of VWF-triggered platelet aggregation on the endothelium subsequently followed by development of TMA. As a consequence, the ratio of VWF antigen level and ADAMTS13 activity was significantly higher in patients with inflammation and sepsis, suggesting that this ratio might be more useful for the diagnosis of highly prothrombotic states including TMA than VWF multimer analysis alone. These findings suggested that ADAMTS13, VWF and related parameters, even in a combined approach, might be useful for the diagnosis and the therapeutic monitoring of patients with sepsis associated thrombotic microangiopathy.

In a case-control study, Molvarec et al (2009) examined if plasma ADAMTS13 activity is decreased in pre-eclampsia. A total of 67 pre-eclamptic patients, 70 healthy pregnant women and 59 healthy non-pregnant women were enrolled in this study. Plasma ADAMTS13 activity was determined with the FRETs-VWF73 assay, while VWF antigen (VWF:Ag) levels with an enzyme-linked immunosorbent assay. The multimeric pattern of VWF was analyzed by SDS-agarose gel electrophoresis. There was no significant difference in plasma ADAMTS13 activity between the pre-eclamptic and the healthy pregnant and non-pregnant groups (median [25 to 75 %]: 98.8 [76.5 to 112.8] %, 96.3 [85.6 to 116.2] % and 91.6 [78.5 to 104.4] %, respectively; $p > 0.05$). However, plasma VWF:Ag levels were significantly higher in pre-eclamptic patients than in healthy pregnant and non-pregnant women (187.1 [145.6 to 243.1] % versus 129.3 [105.1 to 182.8] % and 70.0 [60.2 to 87.3] %, respectively; $p < 0.001$). The multimeric pattern of VWF was normal in each group. Primiparas had lower plasma ADAMTS13 activity than multi-paras (92.6 [75.8 to 110.6] % versus 104.2 [92.1 to 120.8] %; $p = 0.011$). No other relationship was found between clinical characteristics, laboratory parameters and plasma ADAMTS13 activity in either study group. The authors concluded that plasma ADAMTS13 activity is normal in pre-eclampsia despite the increased VWF:Ag levels. However, further studies are needed to determine whether a decrease in plasma ADAMTS13 activity could predispose pre-eclamptic patients to develop HELLP syndrome.

Uemura et al (2010) stated that ADAMTS13 is a metalloproteinase, produced exclusively in hepatic stellate cells, and specifically cleaves highly multimeric VWF, which plays a pivotal role in hemostasis and thrombosis, and its function is dependent on its multimeric state. Deficiency of ADAMTS13 results in accumulation of unusually large VWF multimers (UL-VWFM) in plasma, in turn induces platelet clumping or thrombi under high shear stress, followed by microcirculatory disturbances. Considering that UL-VWFM, the substrate of ADAMTS13, is produced in transformed vascular endothelial cells at sites of liver injury, decreased ADAMTS13 activity may be involved in not only sinusoidal microcirculatory disturbances, but also subsequent progression of liver injuries, eventually leading to multi-organ failure. This concept can be applied to the development or aggravation of liver diseases, including liver cirrhosis, alcoholic hepatitis, veno-occlusive disease, and adverse events after liver transplantation. These results promise to bring further understanding of the pathophysiology of liver diseases, and offer new insight for development of therapeutic strategies.

Okano et al (2010) evaluated changes of plasma ADAMTS13 activity and its clinical relevance in patients with hepatectomy. Plasma ADAMTS13 activity and its related parameters were sequentially determined after hepatectomy in 70 patients. ADAMTS13 activity significantly decreased from pre-operative $67.0 \pm 30.6 \%$ to $48.1 \pm 24.6 \%$ after hepatectomy ($p < 0.0001$). Pringle's maneuver for longer than 45 mins ($p = 0.0007$) and major hepatectomy ($p = 0.0002$) were significantly associated with the decrease of ADAMTS13 activity to less than 40 %. The decreased ADAMTS13 activity reflected post-operative thrombocytopenia ($p = 0.0028$) and hyper-bilirubinemia ($p < 0.05$). The authors concluded that plasma ADAMTS13 activity significantly decreased after hepatectomy due to ischemic injury together with liver mass reduction, reflecting a post-operative liver dysfunction. They stated that monitoring of ADAMTS13 activity may be useful to prevent further development of the liver failure after hepatectomy. Well-designed studies are needed to ascertain the clinical value of ADAMTS13 in monitoring liver diseases.

Choi et al (2011) examined ADAMTS13 activity as well as the ADAMTS13 gene mutation in children with hemolytic uremic syndrome (HUS). A total of 18 patients, including 6 diarrhea-negative (D-HUS) and 12 diarrhea-associated HUS (D+HUS) patients, were evaluated. The extent of VWF degradation was assayed by multimer analysis, and all exons of the ADAMTS13 gene were PCR-amplified using Taq DNA polymerase. The median and range for plasma activity of ADAMTS13 in 6 D-HUS and 12 D+HUS patients were 71.8 % (22.8 to 94.1 %) and 84.9 % (37.9 to 119.9 %), respectively, which were not statistically significantly different from the control group (86.4 %, 34.2 to 112.3 %) ($p > 0.05$). Five ADAMTS13 gene mutations, including 2 novel mutations [1584+2T>A, 3941C>T (S1314L)] and 3 polymorphisms (Q448E, P475S, S903L), were found in 2 D-HUS and 1 D+HUS patients, which were not associated with deficiency of ADAMTS13 activity. Whether these mutations without reduced ADAMTS13 activity are innocent bystanders or predisposing factors in HUS remains unanswered.

Ikeda et al (2011) noted that chronic liver injury evokes a wound healing response, promoting fibrosis and finally hepatocellular carcinoma (HCC), in which hepatic stellate cells play an important role. Although a blood marker of hepatic stellate cells is not known, those cells importantly contribute to the regulation of plasma ADAMTS13 activity. In this study, plasma ADAMTS13 activity was used to predict development of HCC in patients with chronic hepatitis B and C. Plasma ADAMTS13 was evaluated in chronic hepatitis B or C patients with or without HCC. Plasma ADAMTS13 activity significantly correlated with serum aspartate aminotransferase and alanine aminotransferase, liver stiffness value, and aspartate aminotransferase-to-platelet ratio index, irrespective of the presence of HCC, suggesting that it may reflect hepatocellular damage and subsequent wound healing and fibrosis as a result of hepatic stellate cell action. During the 3-year follow-up period for patients without HCC, it developed in 10 among 81 patients. Plasma ADAMTS13 activity was significantly higher in patients with HCC development than in those without and was a significant risk for HCC development by uni-variate and multi-variate analyses. Furthermore, during the 1-year follow-up period for patients with HCC treated with radiofrequency ablation, HCC recurred in 55 among 107 patients. Plasma ADAMTS13 activity or antigen level was significantly higher in patients with HCC recurrence than in those without and was retained as a significant risk for HCC recurrence by multi-variate analysis. The authors concluded that higher plasma ADAMTS13 activity and antigen level was a risk of HCC development in chronic liver disease. They stated that plasma ADAMTS13 as a potential marker of hepatic stellate cells may be useful in the prediction of hepato-carcinogenesis.

In an observational study, Freynhofer et al (2011) investigated the alterations of plasma VWF and ADAMTS13 following cardioversion (CV) and evaluated the predictive value of these parameters for recurrence of atrial fibrillation (AF). These researchers determined plasma levels of VWF and ADAMTS13 in 77 patients before and immediately after CV, as well as 24 hours and 6 weeks thereafter, by means of commercially available assays. The VWF/ADAMTS13-ratio was significantly elevated immediately after CV ($p = 0.02$) and 24 hours after CV ($p = 0.002$) as compared to baseline levels. ADAMTS13, 24 hours after CV, exhibited a significant association with recurrence of AF (hazard ratio [HR]: 0.97; $p = 0.037$). Accordingly, tertiles of ADAMTS13 showed a step-wise inverse correlation with the risk of recurrent AF (HR: 0.50; $p = 0.009$). After adjustment for confounders, ADAMTS13 remained significant as an independent predictor of recurrent AF (HR: 0.61; $p = 0.047$). Similarly, the VWF/ADAMTS13-ratio, 24 hours after CV, was associated with rhythm stability and remained an independent predictor of recurrent AF (HR: 1.88; $p = 0.028$). The regulation of VWF and its cleaving protease ADAMTS13 after CV might play a critical role in producing a pro-thrombotic milieu immediately following CV for AF. The authors concluded that since ADAMTS13 plasma concentration as well as the VWF/ADAMTS13-ratio are

independently associated with rhythm stability, these indexes might be used for prediction of recurrence of AF. These findings need to be validated by well-designed studies.

Habe and co-workers (2012) noted that ADAMTS13, endothelial VWF and related proteins are involved in the pathogenesis of some life-threatening systemic thrombotic coagulopathies. Changes of plasma ADAMTS13 activity in TTP is well-known but is also involved in septic disseminated intravascular coagulation (DIC). These researchers investigated the ADAMTS13 activity, VWF and VWF pro-peptide (VWFpp) antigens in 69 patients with DIC, 143 with non-DIC, 21 with TTP and 23 with atypical HUS (aHUS) for diagnosis of DIC. The plasma ADAMTS13 activity was significantly low in patients with DIC, and the plasma levels of VWF and VWFpp antigens, were the highest in these patients, but there were no significant differences in the plasma VWFpp levels between the patients with DIC and those with aHUS. The difference in the plasma ADAMTS13 activity, the VWF and VWFpp antigens between DIC and non-DIC cases was significant in those with infectious and malignant diseases, but the difference in the VWFpp/ VWF ratio were significant only in subjects with infectious diseases. As an indicator for prognosis, the plasma levels of VWFpp were significantly higher in non-survivors than in survivors. Then, VWFpp/ VWF ratio and VWFpp/ADAMTS13 ratio will be potent informative indicators in DIC. The authors concluded that these findings suggested that ADAMTS13/VWF profiles may have important roles in the pathogenesis of DIC, and that ADAMTS13 and VWFpp are useful indicators for the diagnosis and prognosis of DIC. These findings need to be validated by well-designed studies.

Mazetto et al (2012) stated that increased levels of inflammatory markers and clotting factors have been related to the pathogenesis and prognosis of venous thromboembolism (VTE). In particular, the imbalance between VWF and ADAMTS13 has been described in patients with arterial thrombosis. In this study, a total of 77 patients with previous VTE and 77 matched controls were selected for the evaluation of the inflammatory markers, FVW, ADAMTS 13, and D-dimer. The presences of post-thrombotic syndrome (PTS) and residual vein obstruction (RVO) were also assessed in patients. Serum levels of tumor necrosis factor- α and interleukin-6 were significantly increased in patients compared to controls (median = 2.25 versus 1.59 pg/ml, $p \leq 0.001$; 1.16 versus 0.98 pg/ml, $p = 0.013$, respectively). Plasma levels and activity of VWF (median = 150.25 versus 95.39 U/dL, $p \leq 0.001$; 145.26 % versus 92.39 %, $p \leq 0.001$) and ADAMTS 13 (median = 1088.84 versus 950.80 ng/ml, $p \leq 0.001$; 96.03 versus 83.64 %, $p \leq 0.001$) were also higher in patients. These investigators further analyzed the subgroups of patients with higher risk for VTE recurrence or VTE sequelae, defined as the presence of high D-dimer levels, RVO or PTS. All inflammatory markers were significantly higher in patients with increased D-dimer. The presence of PTS or RVO was not associated with higher inflammatory or coagulation parameters. The increased levels of inflammatory markers and VWF may suggest that there is a persistence of inflammatory activity in patients even at long periods after the VTE episode. In this context, it may be postulated that increased levels of ADAMTS13 could represent a compensatory mechanism against persistently increased levels of VWF. Moreover, increased inflammatory activity was associated with increased D-dimer levels, thus it is possible that this inflammatory activity may also be related to the risk of VTE recurrence.

Sonneveld et al (2014) stated that VWF plays an important role in hemostasis by mediating platelet adhesion and aggregation. Ultra-large VWF multimers are cleaved by ADAMTS13 in smaller, less pro-coagulant forms. An association between high VWF levels and cardiovascular disease has frequently been reported, and more recently also an association has been observed between low ADAMTS13 levels and arterial thrombosis. These investigators reviewed the current literature and performed meta-analyses on the relationship between both VWF and ADAMTS13 with arterial thrombosis. Most studies showed an association between high VWF levels and arterial thrombosis. It remains unclear whether ADAMTS13 is a causal independent risk factor because the association between low ADAMTS13 and arterial thrombosis is so far only shown in case-control studies. The authors concluded that prospective studies are awaited; a causal role for ADAMTS13 is supported by mice studies of cerebral infarction where the infusion of recombinant human ADAMTS13 reduced the infarct size.

Diagnosis of Acute Myelogenous Leukemia

Zhang and colleagues (2014) examined the changes of VWF-cleaving protease (ADAMTS13) activity and VWF antigen (VWF: Ag) level in patients with acute myelogenous leukemia (AML) before and after treatment and evaluated their clinical significance. A total of 73

AML patients were enrolled in this study, the sodium citrate anti-coagulated plasma was collected before and after their induction chemotherapy. Fluorescence resonance energy transfer substrate VWF73 (FRETs-VWF73) assay was established to detect the plasma ADAMTS13 activity while VWF: Ag level was measured by enzyme-linked immunosorbent assay (ELISA). Results showed that the ADAMTS13 activity in newly diagnosed patients with AML before induction therapy was obviously lower than that in normal controls (63.3 ± 25.5) % versus (105.1 ± 37.7) ($p < 0.01$), while the VWF: Ag level was higher than that in normal controls (226.6 ± 127.0) % versus (111.4 ± 39.7) % ($p < 0.01$). After standard induction chemotherapy, the ADAMTS13 activity of AML patients in complete remission period was higher than that in AML patients before therapy ($p < 0.01$), and was not significant difference with that in normal controls; the VWF: Ag was significantly lower than that in AML patients before therapy ($p < 0.01$), but it still was higher than that in controls ($p < 0.05$). The ADAMTS13 activity in newly diagnosed AML patients complicated with infection before therapy was obviously lower than that in AML patients without infection (52.2 ± 20.6) % versus (73.9 ± 24.7) % ($p < 0.01$), while the VWF: Ag level was significantly higher than that in AML patients without infection (262.2 ± 135.7) % versus (193.8 ± 110.2) % ($p < 0.05$). The ADAMTS13 activity in AML patients with disseminated intravascular coagulation (DIC) was significantly lower than that in AML patients without DIC (42.0 ± 14.5) % versus (73.4 ± 22.7) % ($p < 0.01$), while the VWF: Ag level was obviously higher than that in AML patients without DIC (274.2 ± 140.0) % versus (204.7 ± 115.5) % ($p < 0.01$). The authors concluded that the ADAMTS13 activity in newly diagnosed AML patients before induction therapy has been confirmed to be lower and the VWF: Ag level to be higher, especially in AML patients with infection or DIC. They stated that the ADAMTS13 and VWF: Ag may play a role in the pathogenesis of AML and the formation of infection and DIC.

Furthermore, National Comprehensive Cancer Network's clinical practice guideline on "Acute myeloid leukemia" (Version 1.2015) does not mention ADAMTS13 as a management tool.

Diagnosis of Cerebral Infarction

Qu and colleagues (2016) noted that raised levels of VWF and reduced levels of ADAMTS13 activity are associated with thrombosis. These researchers investigated the relationships between plasma levels of VWF and ADAMTS13, their ratios, and the occurrence of cerebral infarction to understand the roles of VWF and ADAMTS13 in cerebral infarction. A total of 94 patients with cerebral infarction and 103 controls were analyzed. Plasma levels of VWF: Ag, VWF ristocetin cofactor activity (VWF: Rcof), and VWF collagen binding activity (VWF: CB) were measured by enzyme-linked immunosorbent assay (ELISA). The ADAMTS13 activity (ADAMTS13) was measured with FRETs-VWF73. The relationship between plasma levels and ratios of VWF and ADAMTS13 and the occurrence of cerebral infarction were analyzed. Patients with cerebral infarction displayed higher VWF: Ag and VWF: Rcof levels and lower ADAMTS13, VWF: CB/VWF: Ag, ADAMTS13/VWF: Ag, and ADAMTS13/VWF: Rcof levels compared to controls ($p < 0.01$). The highest quartiles of VWF: Ag (odds ratio [OR] = 5.11, 95 % confidence interval [CI]: 1.49 to 17.50) and VWF: Rcof (OR = 5.04, 95 % CI: 1.62 to 15.66) and the lowest quartiles of VWF: CB/VWF: Ag (OR = 5.91, 95 % CI: 1.95 to 17.93), ADAMTS13/VWF: Ag (OR = 9.11, 95 % CI: 2.49 to 33.33), and ADAMTS13/VWF: Rcof (OR = 3.73, 95 % CI: 1.39 to 10.03) are associated with cerebral infarction. The authors concluded that an association was found between reduced levels of VWF: CB/VWF: Ag, ADAMTS13/VWF: Ag, and ADAMTS13/VWF: Rcof ratios and cerebral infarction. They stated that these findings suggested that increased levels of VWF and reduced levels of ADAMTS13 activity may contribute to the pathogenesis of cerebral infarction.

Prediction of Excessive Post-Operative Drainage after Coronary Artery Bypass Grafting

Mazur and associates (2014) stated that routine coagulation tests and bleed-scores fail to identify patients at risk of excessive post-operative drainage following coronary artery bypass grafting (CABG). These researchers examined if lower VWF and higher ADAMTS13 are associated with a high post-operative drainage after CABG. In the prospective cohort study, VWF: Ag, VWF:RCO, VWF:CB, ADAMTS13 antigen (ADAMTS13:Ag) and ADAMTS13 activity were measured on the day of elective on-pump CABG in 232 consecutive patients without a prior history of hemorrhagic diathesis, including von Willebrand disease (95 % discontinued aspirin pre-operatively). Post-operative drainage and blood product use were recorded. A comparison of extreme drainage quartiles ($n = 56$) showed that individuals with the highest drainage volumes have mean VWF: RCO lower by 19 % ($p < 0.0001$), median VWF: Ag lower by 19 % ($p < 0.0001$), ADAMTS13: Ag higher by 8 % ($p = 0.0002$), ADAMTS13 activity higher by 9 % ($p = 0.01$) and fibrinogen lower by 14 % ($p =$

0.03) than those with the lowest drainage. Linear regression analysis showed that pre-operative VWF: RCO ($b = -4.83$, $p = 0.002$) and fibrinogen ($b = -61.52$, $p = 0.04$) are the only independent predictors of post-operative drainage. Multi-variate logistic regression demonstrated that pre-operative VWF: RCO in the lowest quartile and ADAMTS13: Ag levels in the highest quartile increased the risk of high (greater than or equal to 1,000 ml) drainage (OR [95 % CI]: 4.88 [1.83 to 13.02], $p = 0.001$ and 3.77 [1.49 to 9.52], $p = 0.005$; respectively). The authors concluded that patients undergoing elective CABG with lower pre-operative VWF: RCO are at risk of having larger post-operative drainage, which suggests a novel contributor to increased peri-operative bleeding in cardiac surgery.

Prediction of Thrombotic Risk in Persons with Systemic Lupus Erythematosus

Martin-Rodriguez et al (2015) noted that severe deficiency of ADAMTS13 activity leads to VWF ultra-large multimers with high affinity for platelets, causing TTP. Other pathological conditions with moderate ADAMTS13 activity exhibit a thrombotic risk. These researchers examined the ADAMTS13 activity in systemic lupus erythematosus (SLE) and its value as a thrombotic biomarker. ADAMTS13 activity, VWF: Ag and multimeric structure, and vascular cell adhesion molecule 1 (VCAM-1) were measured in plasma samples from 50 SLE patients and 50 healthy donors. Disease activity (SLE disease activity index [SLEDAI]) and organ damage (systemic lupus international collaborating clinics) scores, thrombotic events, anti-phospholipid syndrome (APS) and anti-phospholipid antibodies (aPLs) were registered. Patients with SLE showed decreased ADAMTS13 activity and high VWF levels compared with controls (66 ± 27 % versus 101 ± 8 %, $p < 0.01$, and 325 ± 151 % versus 81 ± 14 %, $p < 0.001$); VCAM-1 levels were higher in SLE patients ($p < 0.05$). Considering 3 groups of SLE patients depending on ADAMTS13 activity (greater than 60 %, 60 to 40 % and less than 40%), comparative analysis showed significant association between ADAMTS13 activity and SLEDAI ($p < 0.05$); as well as presence of aPLs ($p < 0.001$), APS ($p < 0.01$) and thrombotic events ($p < 0.01$). Reduced ADAMTS13 activity and increased VWF levels were especially notable in patients with active disease and with aPLs. The authors concluded that ADAMTS13 activity, in combination with other laboratory parameters, could constitute a potential prognostic biomarker of thrombotic risk in SLE.

Diagnosis and Monitoring of Diabetic Retinopathy

Domingueti and colleagues (2016) evaluated the association between plasma levels of VWF, ADAMTS13 and d-Dimer, which consist on endothelial dysfunction and hypercoagulability biomarkers, and cystatin C with retinopathy in type 1 diabetic patients. Patients were classified according to presence ($n = 55$) or absence ($n = 70$) of retinopathy. Plasma levels of VWF, ADAMTS13, d-Dimer and cystatin C were evaluated by ELISA and ADAMTS13 activity was evaluated by FRET. Plasma levels of VWF ($p = 0.033$), ADAMTS13 activity ($p = 0.014$), d-Dimer ($p = 0.002$) and cystatin C ($p < 0.001$) were elevated in diabetic patients with retinopathy compared to those without this complication. The multivariate logistic regression analysis showed that ADAMTS13 activity ($p = 0.031$) d-Dimer ($p = 0.015$) and cystatin C ($p = 0.001$) remained associated with retinopathy after adjustment for age, diabetes duration, use of statin, use of angiotensin-converting enzyme inhibitors (ACEi) or angiotensin antagonist, use of acetylsalicylic acid and glomerular filtration rate. The authors concluded that ADAMTS13 activity, d-Dimer and cystatin C are associated with retinopathy in type 1 diabetic patients and are promising biomarkers for the diagnosis and monitoring of diabetic retinopathy.

Monitoring of Renal Function Following Kidney Transplantation

Mota and associates (2015) stated that kidney transplantation is the key for patients with end-stage renal disease, improving quality of life and longer survival. However, kidney transplantation triggers an intense inflammatory response and alters the hemostatic system, but the pathophysiological mechanisms of these changes are not completely understood. In a cross-sectional, cohort study, these researchers investigated hemostatic biomarkers in Brazilian renal transplanted patients according to renal function and time after transplantation. A total of 159 renal transplanted patients were enrolled and D-Dimer (D-Di), thrombo-modulin (TM), VWF, and ADAMTS13 plasma levels were assessed by ELISA. An increase of D-Di was observed in patients with higher levels of creatinine. ADAMTS13 levels were associated with creatinine plasma levels and D-Di levels with glomerular filtration rate. The authors concluded that these results suggested that D-Di and ADAMTS13 can be promising markers to estimate renal function.

They stated that ADAMTS13 should be investigated throughout the post-transplant time to clarify the participation of this enzyme in glomerular filtration and acceptance or

rejection of the graft in Brazilian transplanted patients.

Acute Cholangitis

Takaya and colleagues (2018) examined the potential role of endotoxemia-related ADAMTS13 in acute cholangitis. A total of 24 patients with acute cholangitis, including 7 with severe acute cholangitis, were recruited in this study. The levels of ADAMTS13:AC, VWF antigen (VWF:Ag), interleukin (IL)-6, IL-8, and tumor necrosis factor (TNF)-alpha in each patient were determined by ELISA, whereas endotoxin (Et) levels were determined by Et activity assay (EAA) analysis. The ADAMTS13:AC and VWF:Ag levels were significantly lower and higher, respectively, in patients with acute cholangitis than in controls. The EAA levels were higher in patients with acute cholangitis than in controls, and were inversely correlated with that of ADAMTS13:AC. Patients with severe acute cholangitis had significantly lower ADAMTS13:AC and higher VWF:Ag levels than those with mild-to-moderate cholangitis. Notably, ADAMTS13:AC was directly correlated with platelet counts and inversely correlated with IL-6 levels, and the VWF:Ag/ADAMTS13:AC ratio was directly correlated with IL-8 and TNF-alpha levels. The authors concluded that imbalance of ADAMTS13:AC and VWF:Ag levels might be associated with severe acute cholangitis, reflecting platelet hyper-aggregability. Severe acute cholangitis has severe pathophysiological features and is complicated by endotoxemia and multiple organ failure.

The authors noted that this was the first report indicating an association between the levels of ADAMTS13:AC and VWF:Ag and those of EAA and cytokines in acute cholangitis. This was a small study (n = 24); its findings need to be validated by well-designed studies.

Acute Pancreatitis

Singh and colleagues (2017) noted that thrombotic micro-angiopathy (TMA) occurring after acute pancreatitis is rarely described. Without prompt intervention, TMA can be, and often is, lethal, so prompt recognition is important. These researchers presented a case of a 61-year old woman with a history of alcohol abuse who presented with epigastric pain, nausea and vomiting after binge drinking. Elevated serum lipase and imaging were suggestive of acute-on-chronic pancreatitis. Although the patient's symptoms of acute pancreatitis subsided, her anemia, thrombocytopenia and acute kidney injury worsened. A peripheral blood smear revealed schistocytes, prompting suspicion for TMA. Therapeutic plasma exchange (TPE) was promptly initiated and she completed 10 TPE sessions that improved her anemia and serum creatinine and resolved the thrombocytopenia. Since TPE was effective and the ADAMTS13 assay revealed 55 % activity in the absence of anti-ADAMTS13 IgG prior to initiation of therapy, a confident diagnosis of TMA caused by acute pancreatitis was made. There was no evidence of relapse 2 years later.

Furthermore, an UpToDate review on "Clinical manifestations and diagnosis of acute pancreatitis" (Vege, 2017) does not mention ADAMTS13 assay.

Melanoma

Weisberg and colleagues (2017) noted that coagulation Factor VIII (FVIII), VWF, and ADAMTS13 play important roles in hemostasis. Patients with cancer are predisposed to thrombosis. However, little is known about the alterations of the genes encoding FVIII, VWF, and ADAMTS13 in patients with cancers. Cross-cancer studies were performed in 25,719 cases from 122 cancer genomic studies. Whole genome sequencing data were analyzed in 345 melanoma cases. All sequencing data and corresponding pathology findings were obtained from the Cancer Genome Atlas (<http://cancergenome.nih.gov/>) and were analyzed using cBioPortal bioinformatics tools (<http://www.cbioportal.org/>). Mutations in FVIII, VWF, and ADAMTS13 were identified in 92 of 122 cancer genomic studies. High mutation rates in FVIII, VWF, and ADAMTS13 (24.2 % to 50 %) were found in patients with skin melanoma from studies of different institutions. Mutations in FVIII, VWF, and ADAMTS13 were also found in other cancer patients, including those with diffuse large B cell lymphoma (22.9 %), lung small cell carcinoma (20.7 %), and colon adenocarcinoma (19.4 %).

Among the 345 melanoma cases (TCGA provisional), FVIII, VWF and ADAMTS13 mutation rates were 15 %, 13 %, and 4 %, respectively. There was a strong tendency towards co-existing mutations of FVIII, VWF, and ADAMTS13 in the same cases. Kaplan-Meier survival analysis indicated that mutations in VWF or ADAMTS13 gene had no effects on patient's overall survival (OS) rate, but mutations in FVIII exhibited a better OS rate (p = 0.0183). The authors concluded that these data suggested that mutations in genes encoding FVIII, VWF, and ADAMTS13 may be associated with melanoma pathogenesis, and that FVIII mutations could

be a useful prognostic marker of melanoma. Moreover, they stated that further studies are needed to understand the biological effect of FVIII, VWF, and ADAMTS13 mutations and confirm the above associations.

Furthermore, National Comprehensive Cancer Network's clinical practice guideline on "Melanoma" (Version 1.2017) does not mention ADAMTS13.

Subarachnoid Hemorrhage

Kumar and colleagues (2017) stated that increased VWF and reduced ADAMTS13 activity are associated with arterial thrombosis. This may also be the culprit mechanism implicated in delayed cerebral ischemia (DCI) after aneurysmal subarachnoid hemorrhage (SAH). In a pilot study, these investigators determine plasma VWF and ADAMTS13 in patients with SAH and healthy subjects; and examined the levels of those markers and outcome after SAH. A total of 40 consecutive patients were enrolled between September 2007 and April 2014.

Plasma samples were collected from SAH patients on post-bleed day (PBD) 0, 1, 3, 5, 7 and 10 and healthy controls; VWF antigen (VWFag) and VWF activity (VWFac) were determined by ELISA and collagen binding assay, respectively. ADAMTS13 activity was determined by the cleavage of a fluorescent substrate. Univariate descriptive statistics and cluster analyses were performed based on outcomes in the group with SAH only. Mean age of SAH patients was 52.4 years (26 to 84 years) and 30 (75 %) were women; 12/40 (30 %) had a high Hunt and Hess grade (IV to V) and 25 (62.5 %) were treated with coil embolization.

Plasma VWFag and VWFac were significantly higher in SAH patients than those in healthy subjects on each PBD ($p < 0.0001$). Concurrently, plasma ADAMTS13 activity in SAH patients was significantly lower than that in healthy subjects ($p < 0.0001$). Among those with SAH, cluster analysis demonstrated that patients with higher VWFag and VWFac and/or lower ADAMTS13 activity might be at risk of increased mortality. The authors concluded that the relative deficiency of plasma ADAMTS13 activity in SAH patients may associate with worse outcome. Moreover, they stated that further larger study is needed to examine if ADAMTS13 supplementation could mitigate microvascular thrombosis and ischemia in patients with SAH; and the effect sizes between different laboratory parameters and mortality outcome provided in this pilot study should facilitate sample size calculations for such future multi-center studies.

This study had several drawbacks:

Diagnosis of Non-Cirrhotic Portal Hypertension

Goel and co-workers (2017) stated that non-cirrhotic intrahepatic portal hypertension (NCIPH) is characterized by TMA of the portal venous system, low ADAMTS13, and high VWF levels. This study aimed to screen for ADAMTS13 mutations, focusing on the CUB domain, in these patients. Prospectively recruited NCIPH patients and healthy volunteers underwent tests for plasma VWF-ADAMTS13 balance. Sanger sequencing of the CUB domain of ADAMTS13 was done in a subset of the NCIPH patients, and the detected mutation was screened for in all the study participants. Next-generation sequencing (NGS) of clinically relevant exome and liver immunostaining for ADAMTS13 was done in patients with detected ADAMTS13 mutation. Plasma VWF-ADAMTS13 balance was significantly altered in 24 NCIPH patients (Child's class A:23, B:1) as compared to 22 controls. On initial sequencing of the CUB domain (17 cases and 3 controls), 1 NCIPH patient showed a rare missense variant (SNV) at position c.3829C >T resulting in p.R1277W (rs14045669).

Subsequent restriction fragment length polymorphism (RFLP) analysis targeted to the R1277W variant did not detect this in any other NCIPH patient, nor in any of the 22 controls. The NCIPH patient with the R1277W variant had severe ADAMTS13 deficiency, consistently high VWF, other missense SNVs in ADAMTS13, VWF, and complement genes.

Immunostaining of his liver biopsy revealed globules of ADAMTS13 within stellate cells. The authors reported missense variants in ADAMTS13, VWF, and complement genes in a patient with NCIPH who had decreased secretion and activity of ADAMTS13 protein.

Moreover, they stated that further studies are needed in NCIPH patients in this regard.

Evaluation of Disease Activity in Inflammatory Bowel Diseases

Cibor and colleagues (2017) evaluated the levels of VWF and ADAMTS13 in inflammatory bowel disease (IBD) and correlated them with the disease activity. Consecutive patients with IBD aged 18 years or older were enrolled in the study. A total of 47 patients with ulcerative colitis (UC), 38 with Crohn's disease (CD), and 50 healthy controls were

included. The white blood cell (WBC) count, hematocrit (Hct), platelet count, fibrinogen, partial activated thromboplastin time, C-reactive protein (CRP), albumin, VWF antigen level (VWF:Ag), VWF ristocetin co-factor activity (VWF:RCo), VWF collagen-binding activity (VWF:CB), and ADAMTS13 antigen level (ADAMTS13:Ag) and activity (ADAMTS13act) were measured. The following ratios were assessed: VWF:RCo/VWF:Ag, VWF:CB/VWF:Ag, VWF:Ag/ADAMTS13act, and ADAMTS13act/ADAMTS13:Ag. Compared to controls, the OR of an elevated VWF: Ag greater than 150 % was 8.7 (95 % CI: 2.7 to 28.1) in the UC group and 16.2 (95 % CI: 4.8 to 54.0) in the CD group. VWF:CB was lower in UC patients, and active CD was associated with a higher VWF: RCo (+38 %). The ORs of VWF:CB/VWF:Ag less than 0.7 (a marker of acquired von Willebrand syndrome) in the UC and CD groups were 11.9 (95 % CI: 4.4 to 32.4) and 13.3 (95 % CI: 4.6 to 38.1), respectively. Active UC was associated with lower ADAMTS13:Ag (-23 %) and ADAMTS13act (-20 %) compared to UC in remission.

Patients with active CD had a 15 % lower ADAMTS13act than controls. The activity of UC, but not that of CD, was inversely correlated with ADAMTS13:Ag ($r = -0.76$) and ADAMTS13act ($r = -0.81$). The authors concluded that complex VWF-ADAMTS13-mediated mechanisms disturbed hemostasis in IBD. A reduced VWF:CB is a risk factor for bleeding, while a lower ADAMTS13 level combined with an elevated VWF:Ag could predispose one to thrombosis.

The authors stated that this study had several drawbacks. First, the groups contained relatively few patients ($n = 47$). Second, the presence of large VWF multimers in plasma was not analyzed. Associations did not necessarily indicate a causal relationship; therefore, in-vitro and animal model studies are needed to elucidate the molecular mechanisms underlying our findings. Finally, this was a case-control study and patients were not followed in terms of thromboembolic events or the duration and severity of bleeding. These researchers stated that if further studies confirm the clinical relevance of these results, they may facilitate the individualization of anti-thrombotic therapy in patients with IBD. The elevated VWF antigen/ADAMTS13 activity ratio indicated an increased risk for thrombo-embolic complications. In this group, the use of anti-coagulation prophylaxis might be considered not only in patients undergoing surgery or hospitalizations due to disease flare-ups but also in out-patients with disease exacerbation. On the other hand, determination of the VWF antigen concentration and VWF activity may facilitate the identification of IBD patients at higher risk for bleeding complications and the management of patients with exacerbated disease, particularly when anti-coagulation prophylaxis is recommended. In this group, a specific for acquired von Willebrand syndrome treatment might be implemented during invasive procedures, especially surgery.

Prediction of Response to Recanalization Therapies in Acute Ischemic Stroke

Bustamante and colleagues (2018) analyzed ADAMTS13 in relation to arterial recanalization in patients treated with intravenous (IV) tissue plasminogen activator (tPA) and in relation to futile recanalization in patients treated with mechanical thrombectomy. Patients with acute ischemic stroke (AIS; $n = 108$) with documented arterial occlusions treated with IV-tPA were selected. ADAMTS13 activity was measured by ELISA in samples collected before treatment. Recanalization was assessed at 2 hours by transcranial Doppler. In 78 consecutive patients treated with endovascular thrombectomy, ADAMTS13 antigen was measured by ELISA and futile recanalization was defined as complete recanalization plus modified Rankin Scale (mRS) score of greater than 2 at 3 months. Independent predictors of recanalization and futile recanalization were determined by logistic regression, adjusted by age, National Institutes of Health (NIH) Stroke Scale score, and time from stroke onset. Patients who achieved tPA-induced recanalization had higher baseline ADAMTS13 activity (78.1 % [68 % to 88 %] versus 70.1 % [61 % to 79 %], $p = 0.021$). In logistic regression analysis, ADAMTS13 activity of greater than 75 % was an independent predictor of recanalization (OR = 6.76 [1.52 to 30.02], $p = 0.012$), together with absence of early ischemic signs and Oxfordshire Community Stroke Project classification. Regarding endovascular therapies, a reduced ADAMTS13 concentration (less than 982 ng/ml) was an independent predictor of futile recanalization (OR = 67.4 [1.4 to 3,282.1], $p = 0.034$), together with age and diabetes mellitus. The addition of ADAMTS13 to clinical predictors of tPA-induced recanalization and futile recanalization improved discrimination and reclassification (integrated discrimination improvement = 10.06 % and 28.4 %, net reclassification improvement = 61.0 % and 107.4 %, respectively). The authors concluded that a reduced ADAMTS13 was associated with poor response to recanalization therapies. These investigators stated that if confirmed in future prospective studies, a panel of blood biomarkers including ADAMTS13 might be a useful tool to guide reperfusion therapies.

Prediction of Survival in Colorectal Cancer

Garam and associates (2018) noted that distant metastasis is a major cause of colorectal cancer-related death, but the mechanism of tumor progression is not fully understood.

There is growing evidence of an interaction between tumor cells and platelets that may influence tumor progression and metastasis formation. Quality and quantity of VWF may regulate the interaction between tumor cells and platelets. These investigators measured the platelet count, VWF antigen (VWF:Ag) levels and ADAMTS13 activity in a large ($n = 232$) cohort of colorectal cancer patients and examined their relationships with the stage of the disease and 5-year survival without thrombotic complications using multi-variable models. Significantly higher platelet counts ($p = 0.005$), VWF:Ag levels ($p = 0.008$) and decreased ADAMTS13 activity ($p = 0.006$) were observed in patients with metastatic disease. Results of the Kaplan-Meier analysis showed that lower platelet counts ($p < 0.0001$), lower VWF:Ag ($p = 0.0008$) levels and higher ADAMTS13 activity ($p < 0.0001$) were associated with better event-free survival (EFS). Finally, to investigate the association between overall EFS and the 3 study variables, multi-variate Cox proportional hazard models were generated. All models were adjusted for age, gender and disease stage. Platelet count, ADAMTS13 activity or VWF:Ag level were incorporated and all of these variables turned out to be age-, gender- and stage-independent predictors of mortality (all HR of greater than 1.7, $p < 0.05$). The authors concluded that this was the first observational study reporting association between higher mortality or thrombotic complications and increased platelet count, increased VWF:Ag levels and decreased ADAMTS13 activity in colorectal cancer.

Moreover, National Comprehensive Cancer Network's clinical practice guidelines on "Colon cancer" (Version 2.2018) and "Rectal cancer" (Version 2.2018) do not mention measurement of ADAMTS13 activity as a management tool.

Prognosis of Traumatic Brain Injury

Russell and associates (2018) stated that decrease of plasma activity of ADAMTS13, a metalloenzyme that cleaves VWF and prevents adhesion and aggregation of platelets, has been reported early after onset of systemic inflammation resulting from infections and after severe trauma. These investigators examined if trauma-induced systemic (sterile) inflammation would be associated with a reduction of plasma ADAMTS13 activity in pediatric patients and its association with disease severity and outcome. Pediatric patients ($n = 106$) with severe trauma at a level 1 pediatric trauma center between 2014 and 2016 were prospectively enrolled. Blood samples were collected upon arrival and at 24 hours and analyzed for plasma levels of ADAMTS13 activity, VWF antigen, collagen binding activity, human neutrophil peptides (HNP) 1-3, coagulation abnormalities, endothelial glycocalyx damage and clinical outcome. Plasma samples were also collected for similar measurements from 52 healthy pediatric controls who underwent elective minor surgery. The median age of patients was 9 years with 81 % sustaining blunt trauma. The median injury severity score was 22 and the mortality rate was 11 %. Plasma levels of ADAMTS13 activity were significantly lower and plasma levels of VWF antigen and HNP 1-3 proteins were significantly higher for pediatric trauma patients on admission and at 24 hours when compared with controls. Finally, the lowest plasma ADAMTS13 activity was found in patients who died from their injuries. The authors concluded that relative plasma deficiency of ADAMTS13 activity may be associated with more severe traumatic injury, significant endothelial glycocalyx damage, coagulation abnormalities and mortality after severe trauma in pediatric patients. These preliminary findings need to be further investigated.

Kumar and colleagues (2019) noted that traumatic microvascular injury (tmVI) is a universal endo-phenotype of traumatic brain injury (TBI) that is responsible for significant neurological morbidity and mortality. The mechanism underlying tmVI is not fully understood. These researchers determined plasma levels of VWF, ADAMTS-13 activity, and human neutrophil peptides (HNP) 1-3 and to correlate these biomarkers with functional outcomes following moderate-severe TBI. A total of 31 consecutive TBI patients (Glasgow Coma Scale [GCS] range of 3 to 12) were enrolled into the study between February 2010 and November 2014. Blood samples were collected on 0, 1, 2, 3, and 5 days after admission and analyzed for plasma levels of VWF antigen (VWF-Ag), collagen-binding activity (VWF-Ac), ADAMTS13 activity, and HNP1-3 proteins. The mean values of plasma VWF-Ag, VWF-Ac, and HNP1-3 were significantly increased in TBI patients compared with those in the healthy controls ($n = 30$). Conversely, the mean plasma values of ADAMTS13 activity in TBI patients were significantly decreased during the first 2 days after admission. This resulted in a dramatic reduction in the ratio of ADAMTS13 activity to VWF-Ag or ADAMTS13

to VWF-Ac in all 5 post-TBI days. Cluster analysis demonstrated that higher median plasma levels of VWF-Ag and HNP1-3 were observed in the cluster with a higher mortality rate. The authors concluded that these findings showed that a relative deficiency of plasma ADAMTS13 activity, resulting from activation of neutrophils and endothelium, may contribute to the formation of microvascular thrombosis and mortality after moderate-severe TBI.

Biomarker for Sorafenib Treatment Efficiency in Hepato-Cellular Carcinoma

Takaya and colleagues (2019) stated that many advanced hepato-cellular carcinoma (HCC) patients are receiving sorafenib treatment. Sorafenib reportedly improves OS significantly in patients with HCC. Prediction of sorafenib response and prognosis in patients with HCC receiving sorafenib treatment are important due to the potentially serious side effects of sorafenib. ADAMTS13 and VWF are associated with the pathophysiology of liver cirrhosis and HCC through their roles in hypercoagulability; they are also associated with angiogenesis via vascular endothelial growth factor (VEGF). The imbalance between ADAMTS13 and VWF was associated with prognosis of various cancers in patients undergoing chemotherapy. These investigators examined ADAMTS13 and VWF as potential biomarkers for sorafenib response and prognosis in patients with HCC receiving sorafenib treatment. A total of 41 patients with HCC receiving sorafenib treatment were included in this study. The initial daily sorafenib dose was 400 mg in all patients. ADAMTS13 activity (ADAMTS13:AC), VWF antigen (VWF:Ag), VEGF levels were determined by ELISA. Uni-variate and multi-variate analyses were used to determine predictive factors for sorafenib response and prognosis in patients with HCC receiving sorafenib treatment. ADAMTS13:AC was significantly higher in patients with stable disease (SD), partial response (PR), and complete response (CR) than in those with progressive disease (PD) ($p < 0.05$). In contrast, VWF:Ag and the VWF:Ag/ADAMTS13:AC ratio were significantly lower in patients with SD, PR, and CR than in those with PD ($p < 0.05$ for both). Multi-variate analysis showed that the VWF:Ag/ADAMTS13:AC ratio was the only predictive factor for sorafenib response and ADAMTS13:AC was the only prognostic factor in patients with HCC receiving sorafenib treatment. The patients with a low ADAMTS13:AC (less than 78.0) had significantly higher VEGF levels than those with a high ADAMTS13:AC (greater than or equal to 78.0) ($p < 0.05$). The authors concluded that VWF:Ag/ADAMTS13:AC ratio and ADAMTS13:AC are potentially useful biomarkers for sorafenib response and prognosis, respectively, in patients with HCC receiving sorafenib treatment. These researchers noted that to their knowledge, this was the 1st report the ADAMTS13-VWF imbalance in association with sorafenib response and prognosis in patients with HCC receiving sorafenib treatment. These preliminary findings need to be validated by well-designed studies.

The authors stated that this study had several drawbacks, including a short observation period and the small sample size ($n = 41$). Cirrhotic patients with HCC occasionally develop thrombosis or inflammation, including portal thrombosis and bacterial over-growth and translocation, which may impact the VWF:Ag/ADAMTS13:AC ratio and ADAMTS13:AC and impact their value as biomarkers. In addition, only 7.3 % of patients were female (male: 38, female: 3). These investigators believed that the difference in gender had no effects in this study because a previous study has reported that the relationships between ADAMTS13:AC and other parameters (e.g., albumin, total bilirubin, aspartate aminotransferase, alkaline phosphatase, and creatinine) were not associated with gender bias.

Biomarker of Delayed Cerebral Ischemia After Subarachnoid Hemorrhage

Jabbarli and colleagues (2020) stated that delayed cerebral ischemia (DCI) is a severe complication of SAH. Clinical and radiographic features of SAH may be helpful in identification of individuals prone to DCI. In a systematic review, these investigators examined the present evidence on predictive value of blood and cerebrospinal fluid (CSF) biomarkers of DCI after SAH. They systematically searched in PubMed, Scopus, Web of Science, and Cochrane Library data-bases for publications before July 15, 2018, reporting correlations between blood/CSF biomarkers and occurrence of DCI and/or vasospasm in SAH patients. Included studies underwent quality assessment according to QUIPS and STARD guidelines. Level of evidence (I to IV) for each of tested biomarkers was evaluated according to GRADE guidelines. Of 2,181 unique records identified in 4 data-bases, 270 original articles and 5 meta-analyses were included to this review. Of 257 blood and CSF parameters analyzed in 16,914 SAH patients, there was no biomarker with positive association with DCI/vasospasm showing level I evidence; 21 biomarkers achieved level II

evidence and could be confirmed as predictive biomarkers. In this review, 6 SNPs (for EET metabolic pathways, COMT, HMGB1, ACE, PAI-1 promoter, and Hp genes) and 15 non-genetic biomarkers (pNF-H, ADAMTS13, NPY, Copeptin, HMGB1, GFAP, periostin, Tau, BNP, NT pro-BNP, hs-TnT, PA-TEGMA, MPV:PLT, NLR, and PLR) were selected as predictive DCI biomarkers. The authors proposed that a panel analysis of the selected genetic and protein biomarker candidates would be needed for further validation in a large SAH cohort.

Biomarker of Early Detection of Hepatocellular Carcinoma

Takaya and colleagues (2020) examined that the VWF to ADAMTS13 ratio as a potential biomarker for early detection of hepato-cellular carcinoma (HCC) in cirrhosis. Serum levels of alpha-fetoprotein, des- γ -carboxy prothrombin, Lens culinaris agglutinin-reactive fraction of alpha-fetoprotein (alpha-fetoprotein-L3%), VEGF, and VEGF receptor-2, as well as the plasma levels of VWF antigen (VWF: Ag) and ADAMTS13 activity (ADAMTS13:AC), were evaluated in 41 cirrhotic patients with HCC undergoing radiofrequency ablation (RFA) and in 20 cirrhotic patients without HCC. The diagnostic accuracy of each biomarker was examined using the receiver operating characteristic curve analysis. The VWF: Ag and VWF: Ag/ADAMTS13:AC ratios were significantly higher in cirrhotic patients with HCC than in those without HCC ($p < 0.05$ and $p < 0.01$, respectively), whereas ADAMTS13:AC was significantly lower in those with HCC than those without HCC ($p < 0.05$). However, no relationship was observed between the VWF: Ag/ADAMTS13:AC ratio and serum tumor markers such as alpha-fetoprotein, des- γ -carboxy prothrombin, and alpha-fetoprotein-L3%. Multi-variate regression analysis identified VWF: Ag/ADAMTS13:AC ratio and alpha-fetoprotein-L3% as significant factors of HCC development. Receiver operating characteristic analysis showed that the VWF: Ag/ADAMTS13:AC ratio and alpha-fetoprotein-L3% had a better performance than alpha-fetoprotein, des- γ -carboxy prothrombin, alpha-fetoprotein-L3%, VEGF, and VEGF receptor-2, VWF: Ag, and ADAMTS13:AC. The VWF: Ag/ADAMTS13:AC ratio was exclusively correlated with tumor volume and stage as well as serum VEGF levels. The authors concluded that the VWF: Ag/ADAMTS13:AC ratio can potentially serve as a novel biomarker for early diagnosis of HCC in cirrhotic patients. Moreover, these researchers stated that further studies need to be performed to validate these findings.

The authors stated that this study had several drawbacks, including the absence of clinicopathologic or prognostic data and the small sample size. Patients with liver cirrhosis sometimes develop thrombosis or inflammation, including portal thrombosis, bacterial overgrowth, and translocation. When the VWF:Ag/ADAMTS13:AC ratio was used as a biomarker for the early detection of HCC, thrombosis and inflammation might affect the value of the ratio. Only patients with hyper-vascular HCCs were analyzed in the present study; thus, pathologically, early or hypo-vascular HCC should be examined in the future.

Biomarker of Monitoring Progression of Acute Myocardial Infarction

Al-Masri and colleagues (2020) noted that coagulation mechanisms and fibrinolytic assembly are important components role players of acute myocardial infarction (AMI) progression. These researchers assessed serial variations in the levels of VWF and ADAMTS13 over the course of AMI and examined their relationship with the cardiovascular risk markers and the patient's clinical characteristics. They studied ADAMTS13, VWF, fibrinogen, and CRP levels in 80 patients with AMI when patients were admitted; post-AMI by 3 to 4 days and at follow-up of 3 months. They compared them with a control group consisting of 36 subjects. AMI had significantly lower levels of ADAMTS13 at AMI and after 3 to 4 days; at follow-up the difference in levels was non-significant, when compared with controls. Similarly, VWF levels were significantly higher in AMI and remained high even at follow-up compared to control subjects. VWF/ADAMTS13 ratio was also significantly higher at AMI and 3 to 4 days while at follow-up difference was non-significant compared to control subjects. Regression analysis between hsCRP and ADAMTS13 showed an inverse relationship ($r = 0.376$, $p < 0.01$), while correlation with VWF was significantly positive ($r = 0.376$, $p < 0.01$), while correlation with VWF was significantly positive. The authors concluded that increased levels of VWF and reduced levels of ADAMTS13 activity may contribute to the pathogenesis of AMI and might prove to be important mediators of AMI progression. These researchers stated that the drawbacks of this study were a relatively small number of samples and short duration. They stated that large-scale, long-term, prospective studies are needed to examine the predictive role of VWF and ADAMTS13 in cardiovascular diseases.

Prediction of Long-Term HBV e Antigen (HBeAg) Sero-Conversion in Individuals with Chronic Hepatitis B

Guo and associates (2019) noted that the relationship between hemostatic system and HBV e antigen (HBeAg) sero-conversion (SC) of chronic hepatitis B (CHB) patients is ill-defined. These researchers examined the predictive value of plasma ADAMTS13 and VWF for CHB patients during 5-year entecavir (ETV) treatment. A total of 114 HBeAg positive CHB patients on continuous ETV treatment were recruited. Liver biopsies were evaluated using the METAVIR scoring system, and plasma ADAMTS13 activity (ADAMTS13: AC) and VWF antigen (VWF: Ag) were determined at baseline, 3, 12, 24, 36, and 60 months, respectively. ETV treatment resulted in an increased ADAMTS13: AC and decreased VWF: Ag (both $p < 0.001$) in CHB patients. Cox multivariate analysis demonstrated that the change of ADAMTS13: AC after 1-year ETV treatment was an independent predictor for HBeAg SC at year 5. The area under the receiver operating characteristic (ROC) curve for the change of ADAMTS13: AC after 1-year ETV treatment plus baseline HBV DNA was 0.873 ($p < 0.001$) to predict SC at year 5. The authors concluded that these findings suggested that increased ADAMTS13: AC after 1 year ETV treatment was associated with a higher SC, and could be used as a surrogate of HBeAg SC in CHB patients during 5-year ETV treatment. Moreover, the presented results raised the possibility of novel supportive therapies of ADAMTS13 supplementation for improving intra-hepatic micro-circulatory disturbance and outcomes of CHB patients, although, more cases should be required and further investigated.

The authors stated that this study had several drawbacks. Even though patients were prospectively followed, it was a retrospective study with a limited number of patients. Moreover, the determination of VWF: Ag concentration was unable to provide specific information regarding the activity of circulating VWF molecular species. A deeper analysis of VWF pro-peptide, VWF multimers and the activity of VWF might better clarify the mechanisms underlying these findings. Furthermore, mutation of the ADAMTS13 gene, marked cytokinemia, enhanced endotoxemia and/or the presence of protease inhibitors may be closely related to declining ADAMTS13: AC14,17. It is necessary to involve these factors into the measurement and analysis of VWF and ADAMTS13 in future study. In addition, It would be recommended to stratify and analyze the relationship between the progression of liver disturbances and ADAMTS13: AC or VWF: Ag in CHB patients during different stage of anti-viral therapy. Also, it will be interesting to determine and analyze the relationship among plasma ADAMTS13, VWF and IP-10 in future study.

Yang and colleagues (2020) stated that ADAMTS13 is a key factor involved in coagulation process and plays a vital role in the progression and prognosis of CHB patients with anti-viral therapy. However, there are few reports about the profile of plasma ADAMTS13 in CHB patients during entecavir maleate (m-ETV) treatment. In this study, a total of 102 HBV e antigen (HBeAg)-positive CHB patients on continuous m-ETV naive for at least 96 weeks were recruited. Patients with liver cirrhosis were excluded using liver biopsies and real-time elastography (RTE). Plasma ADAMTS13 and IL-12 levels were examined at baseline and 12, 24, 48, 72, and 96 weeks, respectively. The change of ADAMTS13 (Δ ADAMTS13) and IL-12 (Δ IL-12) possessed a significant relationship in CHB patients with HBeAg SC at 48-week m-ETV treatment ($p < 0.001$), but no significance in patients without SC. Furthermore, Cox multivariate analysis demonstrated that the change of ADAMTS13 (IL-12) was an independent predictor for HBeAg SC at week 96, and the ROC for the Δ ADAMTS13 (Δ IL-12) in CHB patients with 48-week m-ETV treatment was 0.8204 (0.8354) ($p < 0.001$, both) to predict HBeAg SC at week 96. The authors concluded that these findings suggested that higher increased ADAMTS13 and IL-12 after 48-week m-ETV treatment contributed to an enhanced probability of HBeAg SC, although the mechanism was undetermined. These researchers stated that quantification of ADAMTS13 (IL-12) during m-ETV treatment may help to predict long-term HBeAg SC in CHB patients, although future studies are needed to confirm the role of these cytokines in the pathogenesis of HBeAg SC.

The authors stated that this study had several drawbacks. First, the reduction of ADAMTS13 level was not only a biomarker of disease severity, but also an independent indicator of organ dysfunction and poor prognosis, which implied the key role of ADAMTS13 involved in the pathogenic mechanisms. However, these investigators have not examined which signal pathway ADAMTS13 is involved in and plays the process of HBeAg SC. Second, the plasma ADAMTS13 level was related to liver cirrhosis; the METAVIR scoring system (liver biopsy, F0~F3) is the gold standard; only some patients subjected to liver biopsy, although all patients underwent liver stiffness assessed via RTE, and RTE is non-invasive and accurately stage hepatic fibrosis. Third, the plasma ADAMTS13: activity (AC) and

antigen should be separately measured for getting precise results. In the study, these researchers detected only the ADAMTS13 protein level according to the operational direction, although a previous study reported that both plasma ADAMTS13: AC and antigen levels decreased with increasing severity of cirrhosis. Finally, it still requires further investigation whether the conclusions can be generalized to other cohorts in different countries with different HBV genotype infections, because only the Chinese patient populations were enrolled in this trial.

Risk Factor for Intracranial Aneurysm

Chen and colleagues (2020) stated that the development of intracranial aneurysm (IA) has been linked to genetic factors. These researchers examined the potential role of genes encoding ADAMTS13 in IA development. High-throughput whole-genome and whole-exome sequencing (WES) were used when screening for deleterious SNVs in ADAMTS genes using samples from 20 Han Chinese patients: 19 with familial IA and 1 with sporadic IA. The variant frequencies in these subjects were compared to those in control individuals found in the Genome Aggregation Database. Transcriptome sequencing and methylation sequencing data were retrieved from the Gene Expression Omnibus (GEO) data-base to identify differentially expressed ADAMTS genes and their methylation sites. These investigators predicted the network of interactions among proteins encoded by the overlapping set of ADAMTS genes showing deleterious variants and both differential expression and abnormal methylation in IA. Possible candidate proteins linked to IA were validated using Western blot analysis. The associations between IA and SNVs rs11750568 in ADAMTS2, as well as rs2301612 and rs2285489 in ADAMTS13, were verified using the Sequenom MassArray system on a separate sample set of 595 Han Chinese patients with sporadic IA and 600 control individuals. A total of 16 deleterious variants in 13 ADAMTS genes were identified in these patients, and 7 of these genes overlapped with the genes found to be differentially expressed and differentially methylated in the GEO data-base. Protein-protein interaction analysis predicted that ADAMTS11 was at the center of the 7 genes. ADAMTS11 protein was lower expressed in IA tissue than in the control cerebral artery. Frequencies of the IA-related SNVs rs11750568 in ADAMTS2 and rs2301612 and rs2285489 in ADAMTS13 were not significantly different between sporadic IA patients and controls. The authors concluded that IA was associated with genetic variants, differential expression, and abnormal methylation in ADAMTS genes, ADAMTS11 in particular.

Risk Factor for Thrombosis in Pediatric Congenital Heart Disease

Katneni and colleagues (2020) noted that risk factors contributing to heightened thrombosis in pediatric congenital heart disease (CHD) patients are not fully understood. Among the neonatal CHD population, those presenting with single ventricular physiology are at the highest risk for peri-operative thrombosis. The VWF and ADAMTS13 interactions have emerged as causative risk factors for pediatric stroke and could contribute to heightened thrombosis in CHD neonates. These researchers examined a cohort of children with single ventricle physiology and undergoing cardiac surgery, during which some patients developed thrombosis. In this cohort, these investigators analyzed the relationship of several molecular features of ADAMTS13 with the plasma and activity levels in patients at risk of thrombosis. Furthermore, in light of the natural anti-thrombotic activity of ADAMTS13, these researchers have sequenced the ADAMTS13 gene for each patient and assessed the role of genetic variants in determining the plasma ADAMTS13 levels using a series of in-silico tools including Hidden Markov Models, EVmutation, and Rosetta. Lower ADAMTS13 levels were found in patients that developed thrombosis. A novel in-silico analysis to evaluate haplotype effect of co-occurring variants identified alterations in relative surface area and solvation energy as important contributors. This analysis suggested that beneficial or deleterious effect of a variant can be reasonably predicted by comprehensive analysis of in-silico assessment and in-vitro and/or in-vivo data. The authors concluded that the findings from this study added to the understanding of the role of genetic features of ADAMTS13 in patients at high-risk of thrombosis related to an imbalanced relation between VWF and ADAMTS13.

Furthermore, UpToDate reviews on “Identifying newborns with critical congenital heart disease” (Altman, 2020), and “Diagnosis and initial management of cyanotic heart disease in the newborn” (Geggel, 2020) do not mention measurement of ADAMTS13 as a management option.

ADAMTS13 Genotyping for Prediction of Relapse and Survival Following Allogeneic Hematopoietic Stem Cell Transplantation from Unrelated Donors

Nomoto and colleagues (2019) noted that relapse remains a major obstacle to the survival of patients with hematologic malignancies after allogeneic hematopoietic stem cell transplantation (allo-HSCT). ADAMTS13, which cleaves VWF multimers into less active fragments, is encoded by the ADAMTS13 gene and has a functional single-nucleotide polymorphism (SNP) rs2285489 (C > T). In a retrospective study, these researchers examined if ADAMTS13 rs2285489 affected the transplant outcomes in a cohort of 281 patients who underwent unrelated human leukocyte antigen (HLA)-matched bone marrow transplantation for hematologic malignancies. The recipient ADAMTS13 C/C genotype, which putatively has low inducibility, was associated with an increased relapse rate (HR, 3.12; 95 % CI: 1.25 to 7.77; $p = 0.015$), resulting in a lower disease-free survival (DFS) rate in the patients with a recipient C/C genotype (HR, 1.64; 95 % CI: 1.01 to 2.67; $p = 0.045$). The authors concluded that the findings of this study suggested that the recipient ADAMTS13 genotype predicted the relapse rate and survival outcome following allo-HSCT from unrelated donors. Thus, ADAMTS13 genotyping in transplant recipients may be a useful tool for evaluating the pre-transplantation risks that can form a basis for appropriately tailoring transplantation strategies when combined with other currently known risk factors. If ADAMTS13 genotype information could be used to identify patients prone to relapse in advance, appropriate prophylactic measures and preemptive treatment can be performed before hematologic relapse, which may lead to better survival outcomes. Even treatment alternatives other than allo-HSCT should be considered. Aside from HLA and killer-cell immunoglobulin-like receptors, genetic variations that affect the survival outcomes by influencing relapse are rare. These researchers noted that given the plausible functional roles of this SNP, the current findings suggest the pivotal role of ADAMTS13 in disease progression, which may contribute to the development of novel prophylactic and therapeutic strategies for relapse, such as alterations in the in-vivo regulation of the ADAMTS13 gene. They stated that further studies are needed to determine if these findings can be expanded to other stem cell sources or to HLA-mismatched transplantation and to validate the present data in other ethnic groups.

The authors stated that this study had 3 major drawbacks. First, the functional roles of the ADAMTS13 SNP in relapse following allo-HSCT remain speculative due to the lack of data from the blood samples of allo-HSCT patients and their donors. Likewise, this study did not examine if there was an association between the serum concentrations of ADAMTS13 and the ADAMTS13 SNP. Second, detailed information on sinusoidal obstruction syndrome (SOS)/veno-occlusive syndrome (VOD), such as its incidence and treatment outcomes, was not available in the present study. Furthermore, the effects of the recipient genotypes on the development and treatment outcome of SOS/VOD were unclear, although previous reports have suggested an association of a low ADAMTS13 activity with the development of SOS/VOD. Third, a validation cohort study was not performed to determine the current observations because it was beyond the scope of the study, making this an urgent issue to address in the future.

Biomarker for Disease Severity and Risk of Micro-Thrombosis in Individuals with Coronavirus Disease-2019 (COVID-19)

Mancini and colleagues (2021) noted that severe coronavirus disease 2019 (COVID-19) is characterized by an increased risk of thromboembolic events, with evidence of micro-thrombosis in the lungs of deceased patients. In a cross-sectional study, these investigators examined the mechanism of micro-thrombosis in COVID-19 progression. They evaluated VWF antigen (VWF:Ag), VWF ristocetin-cofactor (VWF:RCo), VWF multimers, VWF pro-peptide (VWFpp), and ADAMTS13 activity in 50 patients stratified according to their admission to 3 different intensity of care units: low (requiring high-flow nasal cannula oxygenation, $n = 14$), intermediate (requiring continuous positive airway pressure [CPAP] devices, $n = 17$), and high (requiring mechanical ventilation, $n = 19$). Median VWF:Ag, VWF:RCo, and VWFpp levels were markedly elevated in COVID-19 patients and increased with intensity of care, with VWF:Ag being 268, 386, and 476 IU/dL; VWF:RCo 216, 334, and 388 IU/dL; and VWFpp 156, 172, and 192 IU/dL in patients at low, intermediate, and high intensity of care, respectively. Conversely, the high-to-low molecular-weight VWF multimers ratios progressively decreased with increasing intensity of care, as well as median ADAMTS13 activity levels, which ranged from 82 IU/dL for patients at low intensity of care to 62 and 55 IU/dL for those at intermediate and high intensity of care. The authors found a significant alteration of the VWF-ADAMTS13 axis in COVID-19 patients, with an elevated VWF:Ag to ADAMTS13 activity ratio that was strongly associated with disease severity. These researchers stated that such an imbalance enhances the hyper-coagulable state of COVID-19 patients and their risk of micro-thrombosis.

The authors stated that this study had several drawbacks. First, the sample size was limited, yet sufficient to identify significant differences across patients at 3 different intensity of care. Second, the study design was cross-sectional, with tests performed on a single sample differentially taken during the course of hospitalization. Studies on larger cohorts of patients followed longitudinally after hospital admission are needed to establish a causal role of the VWF-ADAMTS13 axis in COVID-19 progression and to estimate the risk for poor outcomes associated with each analyzed variable. To-date, it is unclear whether the high levels of VWF are somehow drivers of COVID-19 progression to its most severe forms or an epiphenomenon of the systemic inflammation characterizing the disease. Notwithstanding these drawbacks, these researchers identified new potential markers of disease severity and provided further evidence supporting inflammatory micro-thrombo-genesis in patients with severe COVID-19.

De Jongh and associates (2021) stated that respiratory failure in COVID-19 patients is one of the most frequent causes for referral to the intensive care unit (ICU). A significant percentage of these patients does not survive the infection due to thrombo-embolic complications. In addition, the vascular system appeared to be involved in the pathogenesis. These investigators examined the role of hemostasis and endothelium on the outcome of COVID-19 patients admitted to the ICU. Blood was drawn from 16 ICU COVID-19 patients for hemostatic analysis. Patients were followed-up till discharge ($n = 11$) or death ($n = 5$). Parameters related to both coagulation and fibrinolysis, though disturbed, were not associated with mortality. In contrast, activated VWF was increased and ADAMTS13 levels were decreased by 2-fold in non-survivors compared with survivors. The authors concluded that these findings established the involvement of the VWF-ADAMTS13 axis in the COVID-19 pathogenesis; thus, demonstrating that these plasma proteins appeared to be strong predictors for ICU mortality. These preliminary findings from a small study ($n = 16$) need to be validated by well-designed studies.

Biomarker for Treatment Response in Individuals with Hepatocellular Carcinoma Before the Initiation of Hepatic Arterial Infusion Chemotherapy

Takaya and colleagues (2020) noted that prediction of hepatic arterial infusion chemotherapy (HAIC) treatment response is important for improving the prognosis in patients with hepato-cellular carcinoma (HCC). The progression of HCC is related to hypercoagulability and angiogenesis. It is known that ADAMTS13 and VWF are related to hypercoagulability. Furthermore, a previous study reported that the association between ADAMTS13 and VWF, and angiogenesis via VEGF. Recently, ADAMTS13 and VWF have been associated with the prognosis in patients with various kinds of cancer undergoing chemotherapy. These investigators examined if ADAMTS13 and VWF become useful biomarkers of treatment response in HCC patients before the initiation of HAIC treatment. A total of 72 patients were enrolled in this study. ADAMTS13 activity (ADAMTS13:AC), VWF antigen (VWF:Ag) and VEGF levels were determined via ELISA. Uni-variable and multi-variable analyses were carried out to determine the predictive factors of treatment response in patients with HCC undergoing HAIC treatment. ADAMTS13:AC levels in HCC patients with stable disease (SD) + partial response (PR) of HAIC treatment were significantly higher than those with progressive disease (PD) ($p < 0.05$). In contrast, VWF:Ag/ADAMTS13:AC ratio and VEGF levels in HCC patients with SD + PR were significantly lower than those with PD (both $p < 0.05$). Patients with high VWF:Ag/ADAMTS13:AC ratio (greater than 2.7) had higher VEGF levels than those with low ratio (less than or equal to 2.7). Multi-variable analysis revealed that VWF:Ag/ADAMTS13:AC ratio was a predictive factor of HAIC treatment response. The authors concluded that VWF:Ag/ADAMTS13:AC ratio may become a useful biomarker of treatment response in HCC patients before the initiation of HAIC treatment.

The authors stated that this study had several drawbacks that included a small sample size and short observation. Cirrhotic patients with advanced HCC occasionally developed thrombosis or inflammation (e.g., portal thrombosis, and bacterial over-growth and translocation). When VWF:Ag/ADAMTS13:AC ratio is used as a biomarker of HAIC treatment response, thrombosis and inflammation may affect the values. Furthermore, VWF:Ag/ADAMTS13:AC ratio has high specificity but moderate sensitivity to predict HAIC treatment response; thus, these researchers stated that they should continue to investigate high-sensitivity biomarkers.

Patients with Type 1 Von Willebrand Disease who Plan on Undergoing Major Surgery

UpToDate reviews on “Clinical presentation and diagnosis of von Willebrand disease”

(Rick, 2021a), and “von Willebrand disease (VWD): Treatment of major bleeding and major surgery” (Rick, 2021b) do not mention ADAMTS13 assay as a management tool.

Prediction of Acute Kidney Injury in Patients with COVID-19

Henry and colleagues (2021) stated that severe COVID-19 is often compounded by a prothrombotic state that is associated with poor outcomes. In a prospective, observational study, these investigators examined ADAMTS13 activity, VWF level (VWF:Ag), and the corresponding ADAMTS13 activity/VWF:Ag ratio, in patients with COVID-19 and for associations with disease progression and acute kidney injury (AKI). Patients presenting to the emergency room (ER) with COVID-19 were enrolled in this trial. ADAMTS13 activity and VWF:Ag were measured at index ER visit. The primary endpoint was severe AKI defined by KDIGO stage 2 + 3 criteria, while the secondary endpoint was peak 30-day COVID-19 severity. A total of 52 adult COVID-19 patients were enrolled. These researchers observed that 23.1 % of the cohort had a relative deficiency in ADAMTS13 activity, while 80.8 % had elevated VWF:Ag. The ADAMTS13 activity/VWF:Ag ratio was significantly lower in patients with severe AKI ($p = 0.002$) and those who developed the severe form of COVID-19 ($p = 0.020$). The ADAMTS13 activity/VWF:Ag ratio was negatively correlated with age ($p < 0.001$) and LDH ($p < 0.001$), while positively correlated with hemoglobin ($p = 0.041$). After controlling for confounders, a 1-unit increase in ADAMTS13/VWF:Ag ratio was associated with 20 % decreased odds of severe AKI. The authors concluded that a low ADAMTS13 activity:VWF:Ag ratio at ER presentation was associated with progression to severe COVID-19 disease and severe AKI, with a pattern suggestive of a secondary microangiopathy. Moreover, these researchers stated that further interventional studies should be carried out to evaluate the restoration of ADAMTS13:VWF:Ag ratio in hospitalized patients with COVID-19.

The authors stated that this study was limited by a relatively small sample size ($n = 52$); however, to the best of their knowledge, was the largest study to-date measuring ADAMTS13 and VWF:Ag in patients with COVID-19, and the only study to have examined these values with respect to outcomes. The cohort in this trial did not experience major thrombo-embolic events; thus, these findings should be confirmed in other cohorts with potentially worse outcomes. However, given the timeline of the pandemic, the presence of a high-risk prothrombotic state was well known by the time the 1st wave hit the authors' institution; therefore, appropriate countermeasures were prepared for hospitalized patients, with the majority receiving heparin. Furthermore, not all patients had a pre-COVID SCr value available to define a baseline; thus, it was possible that AKI may be under-represented in this cohort if patients presented with AKI that did not improve during the observation period and were thus classified as unchanged from baseline. Finally, though informative from pathophysiologic standpoint, the clinical value of ADAMTS13:VWF:Ag ratio remains unclear and requires further investigation.

Prediction of Adverse Cardiovascular Outcomes in Individuals Undergoing Percutaneous Coronary Intervention

Tscharre and colleagues (2021) noted that VWF and its cleaving protease ADAMTS13 are pivotal mediators of thrombosis and are associated with the progression of atherosclerosis. These investigators examined the impact of VWF, ADAMTS13 and VWF/ADAMTS13 on long-term major adverse cardiovascular outcomes (MACE) in patients undergoing percutaneous coronary intervention (PCI). They analyzed 701 patients undergoing PCI between 2003 and 2006; VWF and ADAMTS13 antigen levels were measured before PCI. Primary endpoint was MACE, a composite of all-cause mortality, ischemic stroke, or myocardial infarction (MI) during 8 years of follow-up; and secondary endpoint was all-cause mortality. Mean age was 63.8 years, 496 (70.8 %) were men. Acute coronary syndrome (ACS) was diagnosed in 347 (49.5 %) patients, stable coronary artery disease (SCAD) in 354 (50.5 %). During follow-up, 228 (32.5 %) patients experienced MACE, and 161 (23.0 %) died. In ACS patients, VWF was significantly associated with MACE (HR 1.402 (95 % CI: 1.003 to 1.959), $p = 0.048$), whereas ADAMTS13 and VWF/ADAMTS13 had no predictive value. In SCAD, neither VWF, ADAMTS13, nor VWF/ADAMTS13 correlated with MACE. VWF was significantly associated with all-cause death in ACS patients (HR 1.841 (95 % CI: 1.187 to 2.856), $p = 0.006$), but not in SCAD (1.394 (95 % CI: 0.856 to 2.269), $p = 0.181$). ADAMTS13 and VWF/ADAMTS13 were not correlated with ACS and SCAD, respectively. The authors concluded that VWF; but not ADAMTS13 and VWF/ADAMTS13 was associated with MACE and mortality in patients with ACS but not SCAD. This finding highlighted the importance of VWF as an essential marker of risk in patients with ACS.

Prediction of Adverse Outcomes in Patients with Non-Valvular Atrial Fibrillation

Wysokinski and colleagues (2020) stated that VWF elevation correlates with the left atrial blood stasis in non-valvular atrial fibrillation (NVAF); however, the long-term impact of elevated VWF in patients with NVAF is not well established. These researchers examined the impact of VWF and ADAMTS13 in conjunction with echocardiographic (ECG) measures of left atrium blood stasis on clinical outcomes. A total of 414 NVAF prospectively recruited (October 4, 2007 to April 27, 2009) patients were followed for 3 years. VWF antigen, VWF activity, ADAMTS13 activity, and ECG findings were assessed at baseline. Thromboembolism (TE) (stroke/transient ischemic attack (TIA)), MI, or TE of other locations), major bleeding, clinically relevant nonmajor bleeding, and all-cause mortality were evaluated by clinical follow-up, questionnaire, or telephone communication. Among 374 patients (mean age of 63.4 ± 12.7 years; 25 % women) who had complete follow-up data, there were 33 TE in 32 patients (8.6 %), 18 deaths (5.1 %), and 33 bleeding events (21 major bleeding and 12 clinically relevant non-major bleeding) in 25 patients (6.7 %). VWF antigen was predictive of TE in the univariate examination (HR: 1.007, 95 % CI: 1.002 to 1.013, $p = 0.011$) but not in multi-variate analysis. VWF was an independent predictor of all-cause mortality (HR: 1.011, 95 % CI: 1.003 to 1.020, $p = 0.011$) and a composite of TE and all-cause mortality (HR: 1.006, 95 % CI: 1.001 to 1.012, $p = 0.039$) in multi-variate analysis. ADAMTS13 was not predictive of clinical outcomes in multi-variate analysis. In contrast to measures of VWF, ADAMTS13 activity was not an independent predictor of TE, bleeding, or death in multi-variate analysis. ADAMTS13 was regulated in-vivo by the unfolding of the substrate -- the VWF multimer. Brisk laminar blood flow unfolded ultra-large VWF molecules, which allowed rapid proteolysis by ADAMTS13. Under static conditions, proteolysis was retarded 1000-fold; thus, measuring ADAMTS13 activity in-vitro did not reflect flow-mediated conditions in-vivo and, not surprisingly, did not correlate with clinical outcomes. The authors concluded that among patients with NVAF, VWF was an independent predictor of poor outcomes including death and a composite of death and TE. As such, VWF measure may help identify high-risk patients and provide further stratification beyond CHA2DS2-VASc assessment.

Monitoring of Pregnant COVID-19 Women

Grandone et al (2022) stated that thrombotic microangiopathy has been invoked as one of the most important mechanisms of damage in COVID-19 patients. Protease ADAMTS13 is a marker of microangiopathy responsible for controlling von Willebrand multimers size.

VWF/ADAMTS13 ratio has been found impaired in COVID-19 patients outside pregnancy. In an observational study, these researchers examined 90 pregnant women admitted to 2 tertiary academic hospitals in Italy with a laboratory-confirmed diagnosis of SARS-CoV-2 infection. Demographic, clinical information and routine laboratory data were collected at the hospital admission and until discharge. They examined if VWF/ADAMTS13 axis imbalance is a predictor of adverse outcomes. Logistic regression analysis, which controlled for potential confounders, was carried out to examine the association between laboratory parameters and clinical outcomes. Most women (55.6 %) were parae, with median gestational age (GA) at admission of 39 weeks. At hospital admission, 63.3 % were asymptomatic for COVID-19 and 24.4 % showed more than 1 sign or symptom of infection.

Nulliparae with group 0 showed VWF/ADAMTS13 ratios significantly lower than non-0, whereas in multiparae this difference was not observed. Logistic regression showed that ratio VWF to ADAMTS13 was significantly and independently associated with preterm delivery (OR 1.9, 95 % CI: 1.1 to 3.5). The authors concluded that the findings of this study demonstrated an imbalance of VWF/ADAMTS13 axis in pregnant women with COVID-19, leading to a significantly higher and independent risk of preterm delivery. These researchers stated that monitoring these biomarkers might support decision-making process to manage and follow-up pregnancies in this setting.

The authors stated that this study had several drawbacks. First, the observational nature of the study without a control group may have affected the internal validity of the study. These investigators could not exclude that some flaws may adversely have affected data interpretation and the generalizability of findings. However, logistic regression allowed them to control for the confounders that previous studies had highlighted. Second, the limited period of observation; clinical follow-up was available for a very limited period of time (days of hospitalization). Thus, these researchers could not rule out different clinical maternal or fetal outcome in a longer frame-time.

However, the most relevant finding highlighted in this study was the preterm delivery; these investigators paid attention to not look for associations with conditions that could have changed over a longer period of follow-up. Third, these researchers did not

analyze the VWF multimeric pattern in their patients. VWF multimers are relevant markers of endothelial damage, that have been hypothesized to drive micro-thrombosis in COVID-19 patients. On the other hand, this analysis is not widely undertaken by routine laboratories, because of length of test and requirement of specialist equipment. In addition, this test suffers from lack of method standardization, and often variable and subjective results. If on the one hand, the multimers analysis would have allowed more speculations on the pathophysiological mechanisms of the disease, on the other hand it would not have added useful information for the practical management of pregnant women with COVID-19.

Monitoring the Development and Progression of Obstructive Sleep Apnea

Huang et al (2022) stated that obstructive sleep apnea (OSA) is strongly associated with obesity, metabolic diseases, coronary artery disease (CAD), stroke, hypertension, and other disorders. In a cross-sectional study, these researchers examined the relationship between circulating ADAMTS13 levels and the presence of OSA. This trial included a total of 223 patients. These investigators used a powerful high-throughput multiplexed immunobead-based assay to detect circulating levels of ADAMTS13. The associations between circulating ADAMTS13 levels and OSA were evaluated by multi-variate logistic regression analysis. Circulating ADAMTS13 levels were significantly elevated in patients with OSA compared with controls (0.8 versus 2.7 $\mu\text{g/ml}$, respectively, $p < 0.001$). After adjusting for confounding factors, circulating ADAMTS13 levels were significantly independently associated with the presence of OSA (OR = 9.96, 95 % CI: 4.11 to 24.13, $p < 0.001$). Furthermore, circulating ADAMTS13 levels showed discriminatory accuracy in assessing the presence of OSA (area under the curve [AUC]: 0.87, 95 % CI: 0.81 to 0.93, $p < 0.001$). The authors concluded that they demonstrated that circulating ADAMTS13 levels could be used to show the presence of OSA with a high degree of sensitivity and specificity. Moreover, these researchers stated that future studies are needed to highlight the inherent mechanisms of OSA and the predictive value of ADAMTS13 in the outcome of patients with OSA. In addition, they identified circulating ADAMTS13 levels as a novel independent marker of incident OSA. These investigators noted that ADAMTS13 can be a marker to predict the occurrence of OSA; however, further research is needed to confirm this association and to elucidate the biological mechanism underlying this association.

The authors stated that this study had several drawbacks. First, it was a cross-sectional study with a relatively small sample size, which meant that it could only show associations, not causality. Second, these investigators did not examine the effect of continuous positive airway pressure (CPAP) therapy or upper airway surgery on plasma ADAMTS13 expression. Third, potential false-positive results have occurred despite multiple corrections; thus, prospective cohort studies are needed to confirm the variants and associations identified in this trial. Fourth, the controls of this study were enrolled from the Otolaryngological Department of the authors' hospital rather than the general population.

Prediction of Severe Outcomes in Individuals Hospitalized with COVID-19

Schroeder et al (2022) stated that the high heterogeneity in the symptoms and severity of coronavirus disease 2019 (COVID-19) makes it challenging to identify high-risk patients early in the disease. Cardio-metabolic co-morbidities have shown strong associations with COVID-19 severity in epidemiologic studies; thus, cardio-metabolic protein biomarkers may provide predictive insight regarding which patients are most susceptible to severe illness from COVID-19. In a retrospective study, these investigators collected plasma samples from 343 patients hospitalized with COVID-19 during the 1st wave of the pandemic. They measured 92 circulating protein biomarkers previously implicated in cardio-metabolic disease. These researchers carried out proteomic analysis and developed predictive models for severe outcomes. They then used these models to predict the outcomes of out-of-sample patients hospitalized with COVID-19 later in the surge ($n = 194$). These investigators identified a set of 7 protein biomarkers predictive of admission to the ICU and/or death within 28 days of presentation to care. Two of the biomarkers, ADAMTS13 and VEGFD, were associated with a lower risk of ICU/death. The remaining biomarkers, ACE2, IL-1RA, IL6, KIM1, and CTSL1, were associated with higher risk. When used to predict the outcomes of the future, out-of-sample patients, the predictive models built with these protein biomarkers outperformed all models built from standard clinical data, including known COVID-19 risk factors. The authors concluded that these findings suggested that proteomic profiling could inform the early clinical

impression of a patient's likelihood of developing severe COVID-19 outcomes and accelerate the recognition and treatment of high-risk patients. These findings need to be validated by well-designed studies.

The authors stated that this study was limited by the precision with which COVID-19 severity could be captured and COVID-19 related outcomes could be tracked. These researchers used ICU admission and death as proxies for severe illness from COVID-19; however, patients may have died or been admitted to the ICU for reasons independent of their COVID-19 status. Furthermore, patients who died after the 28-day follow-up period or outside the hospital or at other hospitals would not have been captured in this trial, which would have under-estimated the true case-count for the death outcome; and may have biased results towards the null. Another limitation was that the hospital laboratory tests as well as protein biomarkers were not measured at the same time for all patients and the time between symptom onset and blood sample collection was not uniform across all patients. In addition, by only collecting discarded blood samples at a single time-point, these researchers were unable to carry out longitudinal analyses.

Battaglini et al (2022) noted that severe acute respiratory syndrome-coronavirus 2 (SARS-CoV-2) results in a wide spectrum of clinical manifestations, with progression to multi-organ failure in the most severe cases. Several biomarkers can be altered in COVID-19, and they can be associated with diagnosis, prognosis, and outcomes. The most used biomarkers in COVID-19 include several proinflammatory cytokines, neuron-specific enolase (NSE), lactate dehydrogenase (LDH), aspartate transaminase (AST), neutrophil count, neutrophils-to-lymphocytes ratio, troponins, creatine kinase (MB), myoglobin, D-dimer, brain natriuretic peptide (BNP), and its N-terminal pro-hormone (NT-proBNP). Some of these biomarkers can be readily used to predict disease severity, hospitalization, ICU admission, and mortality, while others, such as metabolomic and proteomic analysis (including ADAMTS13), have not yet translated to clinical practice. The authors concluded that novel analytic strategies including metabolomics and proteomics offer interesting insights for early detection of patients at higher risk of severe disease and death; however, their limited availability restricts their widespread clinical use. These researchers stated that further investigations are needed to identify a core set of laboratory biomarkers that can be used in daily clinical practice to predict prognosis and outcome in hospitalized patients with severe COVID-19.

Mohebbi et al (2023) noted that many studies have linked COVID-19 with endothelial dysfunction and reported elevated levels of endothelial biomarkers in this disease. In a systematic review and meta-analysis, these investigators examined available evidence in this respect. They carried out a systematic literature search of PubMed and Scopus databases to identify studies examining biomarkers of endothelial dysfunction in COVID-19 patients. Pooled standardized mean differences (SMDs) and their 95 % CIs were calculated for each biomarker using random effect model. A total of 74 studies with 7,668 patients were included. In comparison to patients with good outcome, those with poor outcome had higher levels of VWF (SMD: 0.83, 95 % CI: 0.59 to 1.07, $p < 0.00001$), VWF:ADAMTS13 (1.23, (0.77 to 1.7), $p < 0.00001$), angiopoietin-2 (Ang-2) (1.06 (0.6 to 1.51), $p < 0.0001$), E-selectin (1.09 (0.55 to 1.63), $p < 0.0001$), P-selectin (0.59 (0.24 to 0.94), $p = 0.001$), syndecan-1 (0.99 (0.6 to 1.37), $p < 0.00001$), mid-regional pro-adrenomedullin (MR-proADM) (1.52 (1.35 to 1.68), $p < 0.00001$), VEGF (0.27 (0.02 to 0.53), $p = 0.03$), soluble fms-like tyrosine kinase-1 (sFLT-1) (1.93 (0.65 to 3.21), $p = 0.03$) and lower levels of ADAMTS13 antigen (-0.69 (-0.9 to -0.47) $p < 0.00001$) and activity (-0.84 (-1.06 to -0.61) $p < 0.0000$). Plasminogen activator inhibitor-1 and tissue plasminogen activator levels were not different between the 2 groups ($p < 0.05$). The authors concluded that there were elevated levels of endothelial dysfunction biomarkers in COVID-19 patients with poor outcome, indicating their possible role in disease severity and prognosis. In particular, MR-proADM, VWF, syndecan-1 and sFLT-1 showed a significant association with poor outcome in these patients.

Determination of the Risk of Endometriosis

Li et al (2023) stated that endometriosis is recognized as a complex gynecological disorder that can cause severe pain and infertility, affecting 6 % to 10 % of all reproductive-aged women. Endometriosis is a condition in which endometrial tissue, which normally lines the inside of the uterus, deposits in other tissues. The etiology and pathogenesis of endometriosis remain ambiguous. Despite debates, it is generally agreed that endometriosis is a chronic inflammatory disease, and patients with endometriosis appear to be in a hyper-coagulable state. The coagulation system plays important roles in hemostasis and inflammatory responses. These researchers employed publicly available

GWAS summary statistics to examine the causal relationship between coagulation factors and the risk of endometriosis. A 2-sample Mendelian randomization (MR) analytic framework was used. A series of quality control procedures were followed in order to select eligible instrumental variables that were strongly associated with the exposures (vWF, ADAMTS13, aPTT, FVIII, FXI, FVII, FX, ETP, PAI-1, protein C, and plasmin). Two independent cohorts of European ancestry with endometriosis GWAS summary statistics were used: U.K. Biobank (4,354 cases and 217,500 controls) and FinnGen (8,288 cases and 68,969 controls). These investigators carried out MR analyses separately in the U.K. Biobank and FinnGen, followed by a meta-analysis. The Cochran's Q test, MR-Egger intercept test, and leave-one-out sensitivity analyses were used to examine the heterogeneities, horizontal pleiotropy, and stabilities of SNPs in endometriosis. The 2-sample MR analysis of 11 coagulation factors in the U.K. Biobank suggested a reliable causal effect of genetically predicted plasma ADAMTS13 level on decreased endometriosis risk. A negative causal effect of ADAMTS13 and a positive causal effect of vWF on endometriosis were observed in the FinnGen. In the meta-analysis, the causal associations remained significant with a strong effect size. The MR analyses also identified potential causal effects of ADAMTS13 and vWF on different sub-phenotypes of endometrioses. The authors concluded that their MR analysis based on GWAS data from large-scale population studies showed the causal associations between ADAMTS13/vWF and the risk of endometriosis. These researchers stated that these findings suggested that these coagulation factors are involved in the development of endometriosis and may represent potential therapeutic targets for the management of this complex disease.

The authors stated that his study had several drawbacks. First, the number of instrumental variables for each coagulation factor ranged from 3 to 13. In addition, some instrumental variables will be discarded during MR analyses when harmonizing the exposure and outcome data. These limitations suggested that the final MR estimates may be subject to influence from the limited number of instrumental variables. Nevertheless, the statistical power calculations for each coagulation factor within each cohort indicated that adequate power was achieved, with estimated statistical power ranging from 80 % to 1. Thus, despite the potential limitations, the results presented in this study remain sufficiently powered to draw robust conclusions. Second, only genome-wide significant SNPs for different coagulation factors were available in the exposure GWAS data, preventing us from performing bi-directional MR analyses. Third, in the context of endometriosis, a female-specific condition, it is noteworthy that existing GWASs examining diverse coagulation factors have been conducted on a sex-combined bias. As 2-sample MR necessitates consistency in the underlying population for both sample sets, it is important to consider potential discrepancies with regard to genetic estimates of coagulation factors in females versus males, which may introduce bias into our MR findings. Fourth, because this study was limited to individuals of European ancestry, the findings may not be generalizable to other populations. These researchers stated that more studies into the causal associations between coagulation factors and endometriosis in other populations are needed.

Ratio of von Willebrand Factor Antigen to ADAMTS13 Activity as a Prognostic Biomarker in Acute-on-Chronic Liver Failure

Takaya et al (2023) noted that acute-on-chronic liver failure (ACLF) has a high risk of short-term mortality. ADAMTS13 is a metalloproteinase that specifically cleaves multimeric VWF. Imbalance between ADAMTS13 and VWF is associated with portal hypertension, which induces ACLF development. A previous study reported that ADAMTS13 activity (ADAMTS13:AC) and VWF antigen (VWF:Ag) are predictive biomarkers of ACLF development in patients with liver cirrhosis (LC). These investigators examined the changes in ADAMTS13:AC and VWF:Ag levels from before to after the development of ACLF to determine their usefulness as a prognostic biomarker in patients with ACLF. A total of 101 patients with LC were enrolled in this study. The level of ADAMTS13:AC and VWF:Ag was determined by ELISA. Cox proportional hazard regression analysis was carried out to determine independent prognostic factors for patients with LC in the post-ACLF group. ADAMTS13:AC levels gradually decreased in the order of non-ACLF group, pre-ACLF group, and finally post-ACLF group. VWF:Ag and the ratio of VWF:Ag to ADAMTS13:AC (VWF:Ag/ADAMTS13:AC) levels gradually increased in the order of non-ACLF group, pre-ACLF group, followed by post-ACLF group. VWF:Ag/ADAMTS13:AC and CLIF-C ACLF scores were associated with prognosis in the post-ACLF group in multi-variate analysis. The cumulative survival of the post-ACLF group was significantly lower for patients with high VWF:Ag/ADAMTS13:AC (greater than 9) compared with those with low VWF:Ag/ADAMTS13:AC (less than or equal to 9) (HR: 10.72, 95 % CI: 1.39 to 82.78, $p < 0.05$). The authors concluded that the VWF:Ag/

ADAMTS13:AC increased according to the progression of ACLF in patients with LC and predicted prognosis in patients with cirrhosis with ACLF.

The authors stated that this study had several drawbacks, including a single-center study and a small sample size ($n = 101$). Furthermore, the occasional occurrence of VTE (e.g., portal thrombosis) in patients with LC may affect the VWF:Ag/ADAMTS13:AC [1,39,40]. A previous study reported that the VWF:Ag/ADAMTS13:AC in patients with severe alcoholic hepatitis was recovered in the recovery stage and worsened in the end-stage compared with patients at the start of the treatment. Furthermore, this study did not examine the change in the VWF:Ag/ADAMTS13:AC after the start of treatment; thus, further studies are needed to examine this issue.

Diagnosis and Prognosis of Portal Vein Thrombosis in Patients with Liver Cirrhosis

Sacco et al (2022) stated that in liver cirrhosis (LC), decreased portal flow velocity, thrombophilia factors, and portal hypertension are considered risk factors for portal vein thrombosis (PVT). In LC, the transformation of the stellate cells causes a progressive decrease of ADAMTS-13, while VWF multimers secretion by endothelial cells is strongly enhanced. This imbalance results in an accumulation of ultra-large VWF multimers that in sinusoidal circulation could favor PVT both in intra- and extra-hepatic branches, mostly in decompensated LC. In a prospective study, these researchers identified possible clinical, biochemical, and hemostatic factors predictive for non-tumoral PVT in a cohort of patients with compensated LC. A total of 79 compensated LC patients were prospectively followed for 48 months, receiving a periodic Doppler-ultrasound (US) liver examination associated with an extensive evaluation of clinical, biochemical, and hemostatic profile. A total of 5 patients developed PVT (cumulative prevalence = 6.3 %), occurring 4 to 36 months after enrollment. In logistic regression analysis, the ADAMTS-13/VWF:GpIbR ratio of less than 0.4 was the only independent variable significantly associated with PVT (OR 14.6, 95 % CI :1.36 to 157.2, $p = 0.027$). A Cox-regression-analysis confirmed this finding (HR = 7.7, $p = 0.027$). The authors concluded that the ADAMTS-13/VWF ratio of less than 0.4 measured in compensated LC could be a reliable predictive biomarker for PVT development, paving the way to novel therapeutic strategies to prevent and treat PVT in this clinical setting.

Suzuki et al (2024) noted that PVT, one of the most prevalent hepatic vascular conditions in patients with LC, is associated with high mortality rates. An imbalance between ADAMTS-13 enzyme and VWF is responsible for hyper-coagulability, including spontaneous thrombus formation in blood vessels. In a retrospective, single-center study, these researchers identified potential prognostic and diagnostic biomarkers in Japanese patients with LC and PVT. A total of 345 patients were divided into 2 groups: 40 patients who developed PVT (PVT group) and 305 who did not develop PVT (NPVT group). Among the 345 patients with LC, 81 % (279/345) were deemed ineligible due to the presence of preventive co-morbidities, active or recent malignancies, and organ dysfunction. The remaining 66 patients were divided into 2 groups: the PVT group ($n = 33$) and the NPVT group ($n = 33$). Plasma ADAMTS-13 activity (ADAMTS-13:AC) and the vWF antigen (VWF:Ag) were measured using ELISAs. Contrast-enhanced, three-dimensional (3D) helical computed tomography (CT) was used to detect and characterize PVT. ADAMTS-13:AC was significantly lower in the PVT group than in the NPVT group. No significant differences in plasma VWF:Ag or liver stiffness were observed between the 2 groups. ADAMTS-13:AC of less than 18.8 was an independent risk factor for PVT on multi-variate analyses (OR: 1.67, 95 % CI: 1.21 to 3.00, $p < 0.002$). The ROC analysis of ADAMTS-13:AC showed an AUC of 0.913 in PVT detection. Patients with PVT having ADAMTS-13:AC of 18.8 or greater ($n = 17$) had higher albumin levels and better prognoses than those with ADAMTS-13:AC of less than 18.8 ($n = 16$). No significant correlations of ADAMTS-13:AC levels with either fibrin degradation product or D-dimer levels were observed. The authors concluded that ADAMTS-13:AC levels could be potential diagnostic and prognostic biomarkers for PVT in Japanese patients with LC. Moreover, these researchers stated that these findings necessitate further investigations to examine the use of ADAMTS-13:AC as a non-invasive diagnostic and prognostic tool and a potential target for future therapeutic interventions.

The authors stated that this study had 2 main drawbacks. First, it was a retrospective, single-center study that involved a small number of Japanese patients with LC ($n = 40$) who either developed or did not develop PVT. Second, a previous study (Sacco et al, 2022) reported the reliability of the ADAMTS-13/VWF ratio as a predictive biomarker for PVT development in patients with LC; however, this trial did not calculate cumulative incidence rates of PVT.

Nie et al (2024) stated that PVT, a complication of LC, is a major public health concern. PVT prediction is the most effective method for PVT diagnosis and treatment. These researchers developed and validated a nomogram and network calculator based on clinical indicators to predict PVT in patients with LC. Patients with LC hospitalized between January 2016 and December 2021 at the First Hospital of Lanzhou University were screened and 643 patients with LC who met the eligibility criteria were retrieved. Following a 1:1 propensity score matching, 572 patients with LC were screened, and relevant clinical data were collected. PVT risk factors were identified using the least absolute shrinkage and selection operator (LASSO) and multi-variate logistic regression analysis. Variance inflation factors and correlation matrix plots were used to analyze multi-collinearity among the variables. A nomogram was constructed to predict the probability of PVT based on independent risk factors for PVT, and its predictive performance was verified using a ROC curve, calibration curves, and decision curve analysis (DCA). Lastly, a network calculator was constructed based on the nomograms. This study enrolled 286 LC patients with PVT and 286 without PVT. LASSO analysis revealed 13 variables as strongly associated with PVT occurrence. Multi-variate logistic regression analysis revealed 9 indicators as independent PVT risk factors, including etiology, ascites, gastro-esophageal varices, platelet count, D-dimer, portal vein diameter, portal vein velocity, aspartate transaminase to neutrophil ratio index, and platelet-to-lymphocyte ratio. LASSO and correlation matrix plot results showed no significant multi-collinearity or correlation among the variables. A nomogram was constructed based on the screened independent risk factors. The nomogram had excellent predictive performance, with an area under the ROC curve of 0.821 and 0.829 in the training and testing groups, respectively. Calibration curves and DCA showed its good clinical performance. Lastly, the optimal cut-off value for the total nomogram score was 0.513. The sensitivity and specificity of the optimal cut-off values were 0.822 and 0.706, respectively. The authors concluded that a nomogram for predicting PVT occurrence was successfully developed and validated, and a network calculator was constructed. This could enable clinicians to rapidly and easily identify high PVT risk groups.

These investigators noted that relevant studies on the novel markers of PVT have also been reported; ADAMTS-13 is expected to be the most promising marker of PVT. ADAMTS-13 activity is inversely correlated with PVT, and is independently associated with PVT in patients with cirrhosis. A recent prospective study (Sacco et al, 2022) showed that an ADAMTS-13/VWF ratio of less than 0.4 in patients with compensated LC could be a reliable biomarker for predicting the development of PVT. The intestinal flora was also closely related to the formation of thrombosis (Hasan et al, 2020). And the portal vein is a bridge between the intestines and liver. The intestinal flora and related metabolites may also be essential for portal vein formation; however, further investigations are needed on the value of these indicators in PVT. These researchers stated that with continuous research and the use of new biotechnologies, an increasing number of novel markers will be discovered, which will be helpful in diagnosing PVT.

Evaluation and Prognosis of Cerebral Small Vessel Disease

In a cross-sectional, single-center study, Sun et al (2021) examined the association between plasma VWF level, ADAMTS13 activity, and neuroimaging features of cerebral small vessel disease (CSVD), including the CSVD neuroimaging markers and the overall CSVD burden. CSVD patients admitted to the authors' hospital from 2016 to 2020 were recruited. Plasma VWF level and ADAMTS13 activity were measured. The overall effect of CSVD on the brain was described as a validated CSVD score. These researchers examined the association between VWF levels, ADAMTS13 activity, and the increasing severity of CSVD score by the logistic regression model. A total of 296 patients were enrolled in this trial. The mean age of the participants was 69.0 years (SD 7.0). The mean VWF level was 1.31 IU/ml, and the ADAMTS13 activity was 88.01 (SD 10.57). In multi-variate regression analysis, lower ADAMTS13 activity and higher VWF level was related to white matter hyperintensity (WMH) ($\beta = -7.31$; 95 % CI: -9.40 to -4.93; $p < 0.01$; $\beta = 0.17$; 95 % CI: 0.11 to 0.23; $p < 0.01$), subcortical infarction (SI) ($\beta = -9.22$; 95 % CI: -11.37 to -7.06; $p < 0.01$; $\beta = 0.21$; 95 % CI: 0.15 to 0.27; $p < 0.01$) independently, but not cerebral micro-bleed (CMB) ($\beta = -2.3$; 95 % CI: -4.95 to 0.05; $p = 0.22$; $\beta = 0.02$; 95 % CI: 0.05 to 0.08; $p = 0.63$). In addition, ADAMTS13 activity was independently negatively correlated with the overall CSVD burden (OR = 21.33; 95 % CI: 17.46 to 54.60; $p < 0.01$) after adjustment for age, history of hypertension, and current smoking. The authors concluded that reducing ADAMTS13 activity change was related to white matter hyperintensity, subcortical infarction, but not with cerebral micro-hemorrhage. Furthermore, ADAMTS13 may have played an essential role in the progression of CSVD.

The authors stated that this study had 2 main drawbacks. First, this study was a cross-sectional, single-center study, with a relatively small number of study subjects ($n = 296$) and the included biological marker indicators were not comprehensive enough. Second, the activity of ADAMTS13 was also demographically statistically different among different races and regions, and the subjects included in this study were all from Shanghai, China; thus, these findings need to be validated in a more extensive prospective study.

Liu et al (2022) stated that CSVD has been found to have a strong association with vascular cognitive impairment (VCI) and functional loss in elderly. Currently, the diagnosis of CSVD mainly relies on brain neuroimaging markers; however, they could not fully reflect the overall picture of the disease. While some biomarkers were found to be related to CSVD, the underlying mechanisms remain unclear. In a systematic review, these investigators examined studies on the progress of biomarkers related to the pathogenesis of CSVD, which is mainly the relationship between these indicators and neuroimaging markers of CSVD. Concerning the pathophysiological mechanism of CSVD, the biomarkers of CSVD have been described as several categories related to sporadic and genetic factors. The authors concluded that monitoring of biomarkers might contribute to the early diagnosis and progression prediction of CSVD; therefore, providing ideas for better diagnosis and treatment of CSVD. However, this review did not address the use of ADAMTS13 as a biomarker of CSVD.

In a community-based, single-center, cross-sectional study, Sun et al (2024) examined if circulating ADAMTS13 activity could offer insights into the mechanism of pathophysiological changes in deep medullary veins (DMVs). This trial was carried out on a community cohort of elderly in Shanghai. Plasma VWF levels and ADAMTS13 activity were measured. A validated DMV score described the overall burden of DMV on the brain. Through ordinal regression models, these investigators examined the correlation between VWF levels, ADAMTS13 activity, and increasing severity of DMV score while adjusting for demographics and cardiovascular risk factors. The study enrolled 262 subjects. The mean VWF level (1.35 ± 0.25) was higher in the DMV group than in the group without DMV (1.25 ± 0.30) ($p = 0.025$), and ADAMTS13 activity (83.76 ± 7.96) was relatively lower. After adjusting for age, sex, alcohol consumption, smoking, hypertension, and diabetes, reduced ADAMTS13 activity ($\beta = -7.78$; 95 % CI: -10.21 to -5.35 , $p < 0.01$) was associated with DMV. Moreover, correlation analysis indicated that ADAMTS13 activity was negatively correlated with the DMV score (Kendall's tau-b = -0.53 , $p < 0.001$). The authors concluded that they study found a negative correlation between ADAMTS13 activity and DMV score, providing some clinical clues for examining the potential pathogenesis of DMV.

These researchers stated that the innovation of this study lied in reporting the correlation between plasma VWF levels, ADAMTS13 activity, and neuroimaging of DMV for the first time; however, this study had several drawbacks. First, the study was a single-center, cross-sectional study with a relatively small sample size ($n = 262$), and the measures of biological markers were not comprehensive enough. Second, ADAMTS13 activity showed a threshold effect, and the increased risk was observed only in the lower range of ADAMTS13. However, the lower range of ADAMTS13 activity in the average population is typically sufficient to maintain the balance of VWF multimer formation and degradation; thus, these findings did not prove that ADAMTS13 activity synergized with VWF levels on DMV. Third, VWF levels were highly variable among individuals and may be influenced by physiological factors, the disease process, or treatment, while ADAMTS13 activity is also affected by race; therefore, these findings need to be validated via a more comprehensive prospective study.

Prediction of Lung Cancer Development

Zhou et al (2025) noted that recent research increasingly highlighted a strong correlation between systemic hyper-coagulability and the risk of lung cancer (LC); however, whether this relationship is causal or merely co-incidental remains unclear. To address this, these researchers carried out a Mendelian randomization (MR) analysis to examine the connections between coagulation factors and LC. Genome-wide association study (GWAS) summary statistics for LC subtypes were retrieved from European ancestry meta-analyses via the IEU Open GWAS project, encompassing 3 distinct histological subtypes: squamous cell carcinoma, adenocarcinoma, and small cell carcinoma. Bi-directional 2-sample MR analyses were carried out using genome-wide significant single nucleotide polymorphisms (SNPs) as instrumental variables. To assess bi-directional causality between 14 coagulation factors and LC, including its subtypes, these researchers used multiple analytical methods, including inverse variance weighting (IVW),

MR-Egger, weighted median, weighted mode, and simple mode. The IVW analysis showed distinct causal associations between coagulation factors and LC subtypes. Elevated plasma levels of factor X (FX) revealed a significant risk effect for squamous cell carcinoma (OR = 1.143, $p = 0.006$), while increased FVIII concentrations were positively associated with adenocarcinoma susceptibility (OR = 1.161, $p = 0.049$). Conversely, higher A disintegrin and metalloproteinase with a thrombospondin type 1 motif, member 13 (ADAMTS13) activity exhibited a protective effect against adenocarcinoma development (OR = 0.781, $p = 0.029$). No statistically significant relationships were identified between any coagulation factors and small cell carcinoma. The authors concluded that leveraging GWAS data derived from large-scale population cohorts, this MR analysis demonstrated significant causal effects of FX, FVIII, and ADAMTS13 on LC risk. They stated that these findings suggested that these coagulation factors were involved in the development of LC and may represent potential therapeutic targets for the management of this complex disease. These preliminary findings need to be validated by well-designed studies.

Systemic Sclerosis

In a cross-sectional study, Ferraz-Amaro et al (2025) examined the relationship between ADAMTS13, systemic sclerosis-scleroderma (SSc), and insulin resistance. This trial included 81 individuals diagnosed with SSc and 76 healthy age- and sex-matched controls. Comprehensive clinical and laboratory characterizations were carried out for both groups. Serum levels of ADAMTS13, standard lipid profiles, and indices of insulin resistance were evaluated in all participants. To determine whether ADAMTS13 serum levels differed significantly between SSc patients and healthy controls (HCs), as well as to examine the relationship between ADAMTS13 values, disease characteristics, and cardiometabolic features, multi-variable linear regression analyses were conducted. Serum ADAMTS13 levels were significantly decreased in patients with SSc compared to HCs after multi-variable analysis. A significant negative correlation was observed between CRP levels and circulating ADAMTS13 levels. However, disease characteristics, including pulmonary, articular, and cutaneous manifestations, did not show significant associations with ADAMTS13 values. Regarding cardiometabolic features, ADAMTS13 levels showed a significant positive association with indices of insulin resistance, a relationship not observed in the control group. The authors concluded that ADAMTS13 serum levels were decreased in patients with SSc. Furthermore a positive correlation existed between ADAMTS13 levels and insulin resistance in SSc.

Furthermore, an UpToDate review on “Clinical manifestations and diagnosis of systemic sclerosis (scleroderma) in adults” (Varga, 2025) does not mention ADAMTS13 as a management tool.

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